SAN BEDA UNIVERSITY
College of Arts and Sciences
Human Kinetics Department

The Effects of Plyometric Training on Selected Collegiate Football Players' Sprint Speed, Agility, and Force Generation

An Undergraduate Thesis presented to
the Faculty of the College of Arts and Sciences
Human Kinetics Department
San Beda University, Manila, Philippines

In partial fulfillment of the requirements in
Research 2

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May 2025

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Acknowledgment

We want to extend our deepest gratitude to all those who have supported and contributed to the success of the completion of this study. This research would not have been possible without the guidance, encouragement, and assistance of numerous individuals and groups, and we are honored to acknowledge their invaluable contributions.

First and foremost, we extend our heartfelt appreciation to our parents, whose unwavering support, love, and encouragement have been the foundation of our academic journey. Their sacrifices and belief in our abilities have inspired us to persevere and strive for excellence.

We also acknowledge ourselves for the dedication, teamwork, and resilience we have shown throughout this research process. From conceptualizing the idea to executing the study and analyzing the results, we have grown not only as researchers but also as individuals committed to advancing knowledge in our field.

Our sincere gratitude goes to the participants of this study, the San Beda University Men's Football Team, whose commitment and effort during the four-week plyometric training intervention were essential to the success of this research. Their willingness to participate and their dedication to improving their athletic performance has been instrumental in generating meaningful findings.

We also extend our appreciation to our fellow researchers from ASM4, whose camaraderie, insights, and shared experiences deepened our understanding and motivated us to push the boundaries of our research. Their collaboration and support have been a source of inspiration throughout this journey.

Special thanks are due to our professors, particularly Sir Greg Emmanuel D. Baniaga, our research supervisor, for his expert guidance, constructive feedback, and

unwavering support. We are also deeply grateful to our research advisors, Sir Ahli Frigillana and Janne Toldoya, for their invaluable advice, encouragement, and expertise. Their insights have helped us navigate the challenges of research and refine our work to meet the highest standards.

We also express our appreciation to our research consultants, Coach Christopher E. Pedimonte, and Sir Alexander Almonte, PTRP. Their professional expertise, practical insights, and endorsements have strengthened the scientific foundation of our study and ensured its relevance to the field of athletic training. Their guidance has been crucial in designing and implementing a plyometric training program tuned to the needs of collegiate football players.

Above all, we give our utmost gratitude to God, whose divine guidance, strength, and wisdom have sustained us throughout this journey of inquiry. From the initial spark of an idea to the execution of the study and the realization of our results, we recognize His hand in every step. His blessings have enabled us to overcome challenges, stay focused, and accomplish this milestone with faith and determination.

This research is a testament to the collective efforts of all those mentioned, and we are deeply grateful for their contributions. Their support has not only made this study possible but has also inspired us to continue pursuing excellence in the field of exercise and sports science.

Abstract

This study looked at how a four-week plyometric training program improved the skills of 25 male football players from San Beda University, aged 18 to 25, in running fast, moving side to side, and jumping high. Plyometric training involves quick, strong movements like jumping and bounding. We tested them before and after with 10-meter, 20-meter sprints, T-Test, Illinois Agility Test, and Vertical Jump Test. They trained twice a week for four weeks, and the results were impressive. Their 10-meter sprint dropped from 2.27 seconds to 1.98 seconds, and the 20-meter sprint went from 3.31 seconds to 2.68 seconds, showing they got faster. Their agility improved too, with the T-Test going from 10.41 seconds to 9.51 seconds and the Illinois Agility Test from 15.22 seconds to 14.11 seconds, meaning they could dodge better. Their jumps rose from 43.76 centimeters to 46.38 centimeters, proving their legs got stronger. Since it worked so well, we created a new six-week program with tougher moves like hurdle jumps and zigzag bounds. This plan builds on their progress to make them even quicker, sharper, and more powerful. In football, speed helps them chase the ball, agility lets them avoid players, and jumping power helps them win headers. This study proves that plyometric training can boost all these skills. It could guide coaches and teams to plan better practices for improving their game.

Keywords: Plyometric training, sprint speed, agility, force generation, collegiate football players, San Beda University, standardized performance tests, T-Test, Illinois Agility Test, Vertical Jump Test, four-week intervention, six-week program, explosive power, football performance, statistical significance

Introduction

Plyometrics is a kind of exercise that focuses on quick and strong movements to boost power. Football is a game where you need to be fast and powerful, so doing plyometric training really helps. Some usual exercises in this training are jump squats, bounding, and depth jumps. These activities are great for making you faster and more powerful, unlike regular strength or weight training that mostly works on building muscle strength instead of speed and power (Markovic & Mikulic, 2020). That is why plyometric training matters a lot for football players. It helps them get better at moving quickly, sprinting faster, and switching directions in a snap. Ghabesh (2020) says that having that explosive power is super important for doing well in sports where you are up against others.

Being quick and agile is a key skill for any footballer to do well in the game. Changing directions fast is something plyometric training can improve, along with speed. Studies by Díaz et al. (2023) and Almoslim & Fadhl (2021) show that adding plyometrics helps athletes boost their agility, quickness, and speed. Plus, this type of training also helps with generating force, which is a big physical perk. Even though plyometrics is used in lots of sports, there's still not enough research on how it directly affects football players' force production, speed, and agility (Huang et al., 2023). Kons et al. (2023) pointed out that since study results aren't always the same, more research is needed. To fill this gap in football, it's worth looking into how plyometric training impacts these skills. Football demands power, speed, and agility to move the ball toward the goal, and this study plans to explore those effects (Díaz et al., 2023). Verkoshansky (2020) mentions that plyometric training started in the Soviet Union back in the 1950s to build explosive power, and as football has changed over time,

training methods need to keep up. This study will tackle the question of whether plyometrics is still helpful for football players today (Moran et al., 2022).

This research will look at how well plyometrics works for sprint speed, agility, and force generation. By doing this, it'll give useful info on how football players perform. Athletes will take part in a four-week plyometric training program with exercises like hurdle jumps, drop jumps, and bounding to build power. This study will check if these exercises really help football players and their main performance areas. It'll include a look at past studies, explain the training program, share the results, and talk about what those findings mean for football training. Finally, it'll give some suggestions and ideas for future studies (Díaz et al., 2023).

Statement of the Problem

This study aims to determine the effects of plyometric training on the sprint speed, agility, and force generation of the selected collegiate football players of San Beda University - Manila, with the help of jump training, bounding, ball throws, and lateral exercises. Specifically, this study aims to answer the following questions.

1. What is the status of the selected collegiate football players of San Beda University - Manila before and after the application of the plyometric training program in terms of standardized performance tests:
    a. 10-meter sprint test;
    b. 20-meter sprint test;
    c. T-test;
    d. Illinois agility test and;
    e. Vertical Jump Test?

2. Is there a significant difference between the standardized performance test results before and after the application of the plyometric training program?

3. What plyometric training program can be developed to enhance sprint speed, agility, and force generation in collegiate football players?

Hypothesis

This study predicts that a four-week plyometric training program, using exercises like jump squats, bounding, medicine ball throws, and lateral plyometric drills, will greatly improve the sprint speed, agility, and force generation of certain college football players from San Beda University in Manila. To be specific, it's expected that: (1) the players' scores in standard tests, such as the 10-meter sprint, 20-meter sprint, T-Test, Illinois Agility Test, and Vertical Jump Test, will get better after training compared to their starting scores, showing boosts in speed, agility, and explosive power; (2) a statistical check will show a clear difference ($p < 0.05$) between the before and after test results for these tests, proving the training works well; and (3) these improvements will back up the idea of creating a plyometric training program designed to sharpen these important skills that football players need to succeed.

Objectives of the Study

The main goal of this study is to find out how a plyometric training program affects the sprint speed, agility, and force generation of selected football players of San Beda University Men's Football Team, specifically:

1. To measure the status of the selected collegiate football players of San Beda University - Manila in terms of sprint speed, agility, and force generation before and after undergoing the 4-week plyometric training program.

2. To analyze the significant difference in the standardized performance test results before and after the application of the plyometric training program
3. To develop a plyometric training program tailored to enhancing sprint speed, agility, and force generation among San Beda football players.

The Significance of the Study

Below, you'll find a list of departments and people who are involved in this issue and how this study will help them.

Athletic Training. This research gives proof that plyometric training works well, which can help college football coaches and trainers at San Beda University improve their methods.

Development of Training Programs. By figuring out which plyometric exercises best boost sprint speed and agility, this study can guide the creation of training plans that fit what athletes really need.

Football Athletes. This study will help football players directly by showing them training techniques that can make them better athletes and stronger competitors on the field.

Future Researchers. The results from this research will fill in missing information about how plyometric training works in football. It will also open doors for more studies on different training styles and how they affect athletic skills.

Philippine Football Federation (PFF). The findings will share useful training ideas with the PFF that can be used at all levels of football, helping the sport grow across the country.

Promotion of Athlete Safety and Injury Prevention. This study can show how certain training methods improve performance while cutting down on injuries by building strength and agility.

San Beda University - Manila, Faculty of Human Kinetics. The results will add to what the Faculty of Human Kinetics knows, offering a plan to include plyometric training in classes and improving training for students.

Scope and Delimitations of the Study

This study will explore how a plyometric training program impacts important skills like sprint speed, agility, and strength in male college football players at San Beda University in Manila. The participants are male players between 18 and 25 years old. Players who are younger, older, or female are not included. The training program consists of two sessions per week for four weeks. Each session has muscle warm-up exercises, general warm-ups, and water breaks to keep things safe and lower the risk of getting hurt. This study uses a quasi-experimental design to carefully look at how plyometric training changes these key skills, apart from the athletes' usual training schedules.

This study would not keep track of what the participants eat or drink, which could affect how they perform, but it is not the focus here. The players' past training or experience levels before starting won't be looked at either. The plyometric training will just be extra work added to their normal daily routine. Also, the team's regular scheduled practices and any personal training they do outside this study might change the results a bit. With these points in mind, the study wants to shed light on how plyometric training helps while pointing out where the findings might have some limits.

Definition of Terms

Agility. In this study, it means being able to move fast and smoothly.

Bounding. In this study, it's a plyometric exercise with big, strong jumps to help improve speed and agility.

Change direction. In this study, it refers to the body turning or moving the opposite way.

Depth jumps. In this study, it's an exercise where you step off a box, land, and then jump up as hard and fast as you can.

Drop jumps. In this study, it's a plyometric move to boost your vertical jump and make you better at leaping.

Football. In this study, it's a sport with two teams of 11 players trying to get the ball into the other team's goal without using hands or arms.

Force generation. In this study, it's about how much force you can push into the ground, like when you sprint or jump.

Hurdle jumps. In this study, it's a plyometric exercise where you jump over obstacles one after another to build explosive power and agility.

Illinois agility test. In this study, it's a standard test to check how agile an athlete is and how fast they can switch directions. It uses a course marked with cones that you run through as quickly as possible while staying in control.

Jump squats. In this study, it's a plyometric exercise to increase lower body power and strength by doing a squat and then jumping up fast.

Lateral plyometric exercises. In this study, these work on strengthening muscles for side-to-side moves, like lateral bounds, jumps, and shuffles, to improve balance, stability, and quick turns.

Medicine ball throws. In this study, it's an exercise where you throw a heavy ball with full effort in different ways, like overhead or chest passes, to build upper body strength, core stability, and power.

Muscle activation. In this study, it means getting your muscles ready to perform well during exercise.

Plyometrics. In this study, it's a type of workout with fast, explosive actions like jumping to improve muscle strength and speed.

Power. In this study, it's how fast you can create force, like being explosive when you start sprinting.

Sprint speed. In this study, it's how fast you can run over a short distance.

Stretch-Shortening Cycle (SSC). In this study, it's when your muscles stretch and then tighten up quickly to make power.

T-test. In this study, it's an agility test that checks how fast you can change directions by running in a T-shaped pattern and timing how long it takes.

Training frequency. In this study, it's how many times you train each week, usually two or three times for plyometrics.

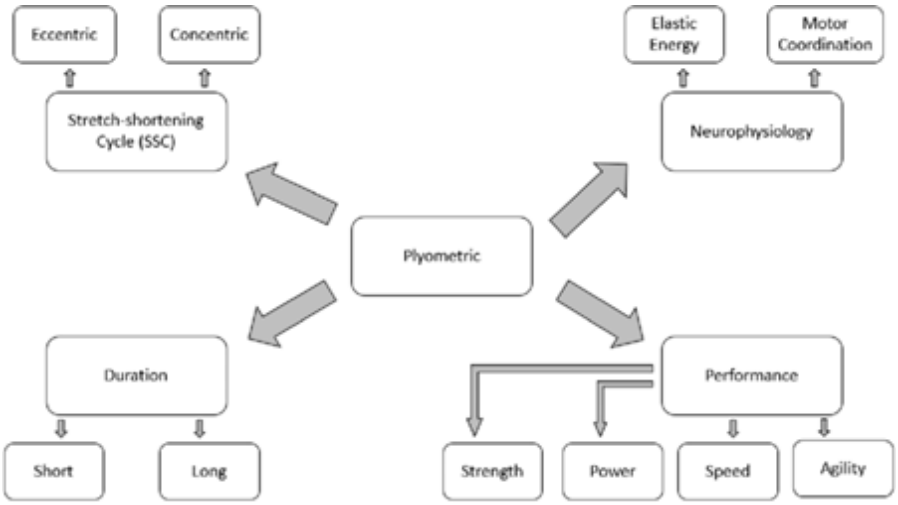
Vertical jump test. In this study, it measures how high an athlete can jump straight up to test their leg power and strength, usually by reaching up with one hand from a standing start.

Warm-up. In this study, it's a set of light exercises done before harder ones to prepare your body and prevent injuries.

Research Framework

Theoretical Framework

Figure 1: Plyometric Training Diagram



Source:https://www.researchgate.net/publication/351108471_Effects_of_Plyometric_Training_on_Speed_and_Agility_among_Recreational_Football_Players

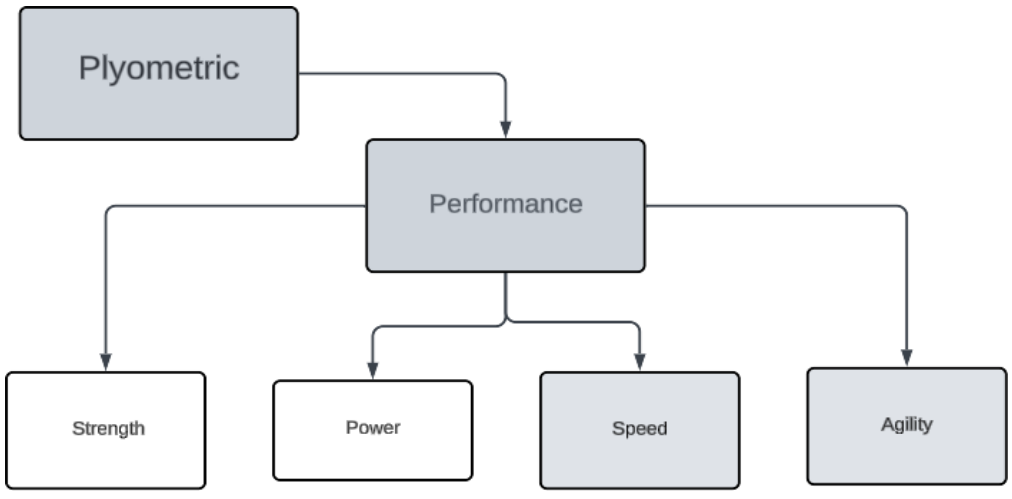
This study wants to figure out how plyometric exercises affect sprint speed, agility, and force generation. It's important because not much research has been done on how these exercises help college football players specifically, even though they've worked well in other sports. The big question this study asks is: How do plyometrics improve sprint speed, agility, and force generation? To answer that, it breaks down into smaller questions about how these exercises connect to sprinting, how they make agility better, and how they build force generation. The idea is that plyometric training uses something called the stretch-shortening cycle in muscles, which makes them tighten up

fast and get more powerful with moves like jump squats and bounding (Hammami et al., 2023). When you do plyometrics, it builds explosive strength by working the muscles and using fast-twitch fibers, which are key for quick moves that help with sprint speed and force. Along with that, the theory about agility and coordination shows how plyometrics can improve those skills, and this can be checked with tests like the T-test. Finally, the force generation theory supports the idea that these exercises boost explosive strength and power, which are super important for athletes to do well (Sáez de Villarreal et al., 2020).

This plan lays out how plyometric training, especially with exercises like jump squats, can improve key skills like speed, agility, and force generation by activating muscles and setting up the groundwork for better agility.

Conceptual Framework

Figure 2: Plyometric Concept

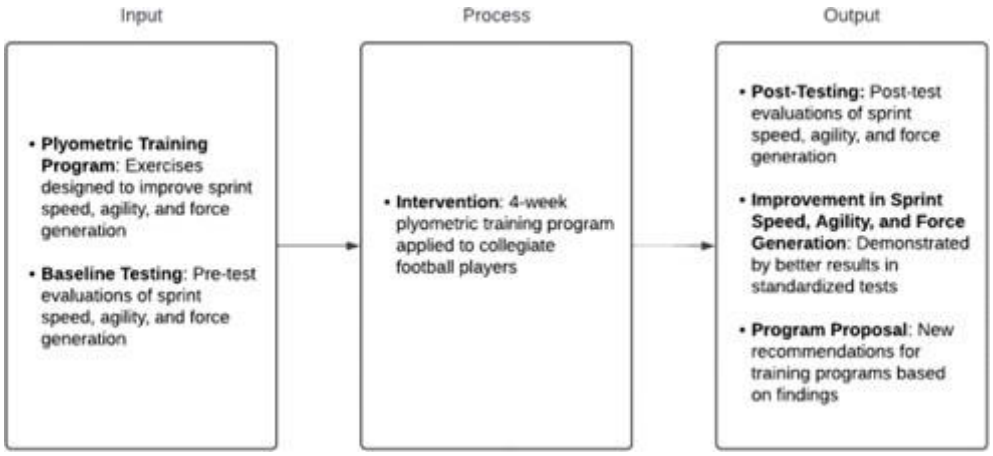


The main point of this framework is to figure out how plyometrics connects to improving key skills like speed, agility, and force generation. Even though studies have shown that plyometric training works well in other sports, there's still not enough information about how it helps football players. Using this framework and plyometric training, we put the theory into action to boost speed, strength, and power. This helps players run faster, move quickly on their feet, and get stronger when controlling the ball, while also adding more knowledge to this topic. Plus, force generation, which is a big deal for explosive actions like tackling and jumping, is something this study checks to see if it improves those key skills.

By exploring these ideas, this framework shows how plyometric training can make athletes better overall by improving their sprint speed, agility, and force production. This study sets up a starting point to see how a plyometric training program can really benefit football players specifically.

Operational Framework

Figure 3: Research Operational Process



This study is split into three parts: input, process, and output. The input part gets everything ready for the research. In this stage, we set up a Plyometric Training Program, which is a series of exercises meant to boost key physical skills in college football players, like sprint speed, agility, and force production. Also, we do baseline testing to measure any improvements accurately. This first test checks each player's starting performance, giving us something to compare to later.

The process part is where most of the action happens. Here, the athletes go through a four-week plyometric training program designed to build those skills. This set time of medium-tough training is super important because it lets us watch and track how their physical abilities get better. By keeping the training period short and controlled, the study can really focus on how plyometric exercises help improve these must-have athletic skills.

The output is the last step, where we look at what the training program achieved. After the four weeks, the players get tested again to see if their sprint speed, agility, and force generation improved. The results from these post-tests will show some progress in those areas,

proving how well the plyometric training worked. If the outcome looks good, it'll contribute to a Program Proposal. This will include new training tips that can help other coaches and trainers who want to use similar programs to boost their athletes' physical performance.

Review of Related Literature and Studies

Plyometric training involves jumping exercises that use just your body weight, and something called the stretch-shortening cycle in your muscles. This cycle helps your nerves and muscles work together to create a lot of force fast, making plyometric exercises a great way to connect strength and speed (Markovic & Mikulic, 2010). That's why it's a key part of training for lots of sports.

Studies show that plyometric training, which started to get better at jumping, also helps in team sports like football. It boosts things like agility, quick changes in direction, sprint speed, and even endurance (Faigenbaum, Myer, & Lloyd, 2021). Plus, it's not just for improving skills, it's also used to help athletes bounce back from injuries.

This review looks at what research says about plyometric training and how it improves physical fitness and important skills for athletes. It digs into the science of how plyometrics work, the different kinds of exercises, and how they make athletic abilities better. The review also checks out recent studies to see how they've helped us learn more about plyometric training and how it can be used to help athletes perform and move better in real situations.

Plyometric Training

Definition and Characteristics of Plyometric Training

Plyometric training (PT) is a type of exercise where muscles stretch and then tighten up fast to build explosive speed and power (Markovic & Mikulic, 2020). It uses the stretch-shortening cycle, which is just a quick stretch of the muscles followed by a sudden squeeze. This process helps muscles work better and make more force, so plyometric exercises are useful for athletes who need quick, powerful moves in their sports (Komi, 2020).

Plyometric training stands out because it involves tough workouts, quick breaks, and a focus on creating as much power as possible (Haff & Nimphius, 2021). It includes different moves like jumps, hops, and bounds that can be adjusted for each athlete. This flexibility makes it a big part of strength and conditioning plans in lots of sports, including football.

Types of Plyometric Training

Traditional Plyometric Training uses exercises like squat jumps, box jumps, and depth jumps. These focus on building explosive power by doing fast movements over and over (Drazan et al., 2020). This kind of plyometrics is good for creating the basic strength athletes need to perform well.

Loaded Plyometric Training adds extra weight, like wearing a weighted vest or using resistance bands, to make the exercises tougher (Beato et al., 2021). This style helps improve both power and strength since athletes must push harder during their moves. It's helpful for athletes looking to level up their overall physical skills.

Unilateral Plyometric Training works on one leg at a time with moves like single-leg hops and bounds, unlike bilateral exercises that use both legs (Meylan & Malatesta, 2021). This method is great for fixing

uneven muscle strength and making one leg stronger, which matters a lot in sports like football where players often use one leg for quick, powerful actions.

Training Parameters: Duration, Frequency, Intensity, and Type of Exercise

Good plyometric training programs need to think about a few important things:

Duration: Plyometric training sessions usually take 20 to 40 minutes. Each exercise should last around 5 to 30 seconds so athletes can put in their best effort (Helgerud et al., 2021). For college athletes, keeping sessions short enough is key so they can stay strong and not get too worn out.

Frequency: Athletes should do plyometric exercises two or three times a week. This gives them enough rest between sessions, which is super important for getting the best results, especially for explosive and powerful moves (Hoshikawa et al., 2020).

Intensity: Plyometric exercises can be easy or hard depending on the move. Something like skipping is light, while depth jumps are tough (Beato et al., 2021). As athletes get better at it, they should start with easier ones and work up to harder ones, so planning carefully matters.

Type of Exercises: The exercises picked should match what the sport needs. For football players, it's smart to include moves that work on side-to-side motion, speeding up, and slowing down to help them play better on the field (Zaharia, 2022).

Periodization Context of Plyometric Training

Plyometric training is strategically integrated into an athlete's periodization plan to optimize performance while minimizing injury risk. Periodization involves structuring training into macrocycles (long-term, typically a year), mesocycles (medium-term, spanning weeks to months), and microcycles (short-term, typically a week) to build athletic capabilities (Bompa & Buzzichelli, 2021). In the context of collegiate football, the 4-week plyometric training program implemented in this study aligns with the pre-season to early in-season timeline, specifically within the preparatory and competitive phases of the macrocycle. During the pre-season (general preparatory phase), the focus is on building foundational strength and power, which corresponds to the initial two weeks of foundational movements in this study's program (e.g., squat jumps, bounding). This phase transitions into the specific preparatory phase, where exercises like depth jumps and lateral bounds enhance sport-specific explosiveness, preparing athletes for the competitive in-season phase (Haff & Triplett, 2022). Within the mesocycle, the 4-week program constitutes a strength-power development block, emphasizing explosive power and agility, critical for football performance. Each microcycle (weekly plan) includes two sessions, balancing intensity and recovery to promote adaptation while preventing overtraining (Chaabene et al., 2024). This periodization approach ensures that plyometric training enhances sprint speed, agility, and force generation in alignment with the demands of the football season.

Specificity of Training Regarding Different Sports

Plyometric training can be adjusted to fit the special needs of different sports. The idea of specificity means that exercises should look like the moves used in the sport to really work well. Football players need explosive side movements, quick sprints, and high jumps during games, so their training should include those too (Kargarfard et al., 2020). Hoff et al. (2021) found that studies show exercises made just for football can lead to big improvements in sprint speed and agility, especially when looking at how college athletes perform before and after training.

Meylan & Malatesta (2021) explained that basketball players often work on exercises to boost their vertical jumps for better rebounding. Meanwhile, track athletes use plyometrics for sprinting to get faster starts and pick up speed. By tweaking plyometric training to match each sport's needs, coaches can help athletes get better at what they do. Moreover, the effectiveness of plyometric training is maximized when integrated into a periodized plan, particularly during the pre-season to early in-season phases, where strength-power mesocycles prepare athletes for competitive demands (Haff & Triplett, 2022).

Plyometric training is all about building explosive power, speed, and agility, which are super important for football players to have. To meet each player's needs, different plyometric exercises can be used, like box jumps, loaded plyometrics with extra weight, and unilateral moves that focus on one leg. The kind of exercise, how long it lasts, how often it's done, and how hard it is are all key parts of making plyometric training work. Sessions that run 20 to 40 minutes, happening two or three times a

week, let players push hard while getting enough rest. Training those copies football moves, like side sprints and jumps, can really help players do better on the field.

Age Differences in Plyometric Training

Eather et al. (2021) found that due to younger athletes' growing body and mind, their muscles and nerves are more adaptive and receptive to training. In comparison, as athletes age, their bodies grow less flexible, limiting the amount of improvement that means younger athletes gain more strength and power from plyometric activities than older ones. Although this study focuses on collegiate players, it's worth considering that younger athletes usually receive significant outcomes from plyometric exercise because of their continued physical growth.

Collegiate athletes can benefit considerably from plyometric activities in connection to they are still young, resulting in increases in strength and power. On the other hand, these improvements are not as pronounced as those seen in younger athletes with rapidly adapting systems. This suggests that plyometric training remains effective at the collegiate level, even if the results are slightly moderate.

Neurophysiological and Mechanical Models Explaining Plyometric Effects

Plyometric training helps athletes perform better by strengthening both their nervous and muscular systems. The neurophysiological model explains how this type of training enhances the nervous system, leading to improved muscle coordination and faster force production. When performing plyometric exercises, the nervous system adjusts by activating more motor units (which are muscle fibers controlled by nerves) and

increasing their firing rate, making muscle contractions more efficient (Graham-Smith et al., 2021). According to Laffaye et al. (2022), studies have shown that collegiate athletes who completed a four-week plyometric program demonstrated noticeable improvements in muscle activation during explosive movements. This is important for increasing sprint speed and agility.

On the other hand, the mechanical model focuses on how plyometric training enhances force production when an athlete pushes against the ground. Research by Kernozek et al. (2020) found that exercises like depth jumps and box jumps help athletes generate more force when preparing to jump, which benefits both jumping and sprinting performance. In a study by Kargarfard et al. (2021), college football players who followed a structured plyometric training program showed significant improvements in their vertical jump height and sprint test results.

Physical Performance Outcomes

Research indicates that college athletes who consistently include plyometric training in their workouts experience significant improvements in lower body strength and explosive power. Beato et al. (2022) found that incorporating plyometric exercises can enhance vertical jump height and power output by 10% to 20%. This type of training is especially beneficial for football players, as it helps develop the strength and explosiveness needed for high-intensity actions like sprinting and jumping. In another study, football players who completed a four-week plyometric training program showed notable improvements in their vertical jump performance. This suggests that plyometric

training not only strengthens muscles but also enhances explosiveness and power, which are essential in football (Hoff et al., 2020).

As a result, athletes who engage in plyometric training consistently can significantly increase their lower body strength and power, improving their overall effectiveness in games. One of the key benefits of plyometric training is its ability to enhance vertical jump height. Football players can see noticeable improvements in their jumping ability after just four weeks of structured plyometric exercises. These improvements are particularly valuable for actions like jumping to head the ball or making quick, powerful movements on the field.

Athletic Expertise Levels

Sato et al. (2022) found that young athletes benefit greatly from plyometric training due to their adaptable muscles and nervous systems. However, collegiate and professional athletes require more specialized training programs to achieve greater improvements. Research by Zeppieri et al. (2021) revealed that college athletes who performed advanced plyometric exercises experienced significant gains in strength and speed, whereas professional athletes often focused on sport-specific drills to refine their performance.

Since this study focuses on collegiate athletes, it suggests that these individuals develop a strong foundation in strength and movement skills. Therefore, they require tailored plyometric programs to further enhance their athletic performance. Additionally, the study highlighted the importance of examining how different playing positions among collegiate athletes influence the effectiveness of plyometric training.

Psychological Factors

Psychological factors, such as motivation and confidence, play a crucial role in how well athletes respond to plyometric training. O'Connor et al. (2020) found that athletes with higher levels of motivation tend to show greater improvements, as they are more willing to engage in training programs and push their limits during workouts. Similarly, research by Lentz et al. (2022) observed that self-confident athletes often perform better in training sessions. This highlights the importance of creating a positive and supportive training environment, as it can significantly boost motivation and enhance overall athletic performance.

Training Adaptation Models

The Stretch-Shortening Cycle (SSC) and Its Implications for Power Development

The stretch-shortening cycle is a key principle in plyometric training. It refers to the muscle's ability to store elastic energy when stretched and released during contraction, involving the eccentric (lengthening) and concentric (shortening) phases. For collegiate athletes, understanding this cycle is essential, as it serves as the foundation for improvements in sprint speed and agility. Seitz et al. (2020) found that athletes who trained with a focus on this cycle experienced a 12% increase in vertical jump height and a 6% improvement in sprint speed.

The biomechanics of the stretch-shortening cycle play a vital role in enhancing force production, particularly in movements like jumping and sprinting, which are crucial in football during both training and competition. Analyzing muscle tightness during specific movements,

along with joint angles, helps in assessing how this cycle influences jump height and how an athlete's power output improves with training.

Sports-Specific Context

The physical demands of each sport play a crucial role in designing an effective training program. McMahon et al. (2020) emphasized that to maximize the benefits of training, exercises should align with the specific physical requirements of the sport. For example, football often requires lateral movements in various game situations, so training programs should be designed to replicate these actions. Similarly, Zourbanos et al. (2023) stated that for athletes to see significant improvements, their supplementary training must match the demands of their sport. Supporting this approach, Paterno et al. (2021) highlighted that understanding the unique challenges of a sport allows coaches to create training programs that better prepare athletes for competition.

Considering the physical demands of football is essential in helping athletes apply their training to real-game situations, allowing them to perform more effectively. By designing training programs that focus on the specific needs of football, this research aims to assist collegiate players in enhancing their sprint speed, agility, and force generation, ultimately improving their overall performance on the field.

Interaction Between Physical Abilities and Technical Skill Performance in Sports

To compete at the highest level in a sport that demands speed, agility, and coordination, such as football, the relationship between

physical abilities and technical skills is crucial. Physical attributes directly influence an athlete's technical performance, especially in a game that involves constant contact with opponents, teammates, and the ball. Haff et al. (2021) concluded that lower body power, as measured by vertical jump height, plays a significant role in improving a collegiate football player's kicking distance.

Similarly, Kargarfard et al. (2020) found that agility and coordination improvements in college football players after plyometric training were key factors in executing technical skills more effectively. This suggests that developing a strong foundation in plyometric exercises can enhance both physical and technical abilities, allowing athletes to perform better in high-level competitions.

Role of Integrated Training Approaches

Integrating plyometric exercises with other forms of training, such as resistance training, has gained popularity due to its positive impact on athletic performance. This approach acknowledges that combining different training methods can lead to better results compared to relying on just one. Ronnestad and Mujika (2021) found that athletes develop both physical abilities and technical skills more effectively when plyometric and resistance exercises are structured to meet the specific demands of their sport. Similarly, Beato et al. (2021) observed that athletes who combined plyometric and resistance training showed greater improvements in sprint speed and vertical jump height compared to those who only performed resistance training. The researchers

attributed these enhancements to the combined effect of both training methods in developing explosive strength and power.

Systematic Review and Meta-Analysis Frameworks

Importance of Quasi-Experimental Research Trial Framework and Quality Assessment

In many cases, randomly assigning participants to groups and conducting controlled experiments can be challenging. As a result, quasi-experimental approaches are often used to examine the potential effects of plyometric training. This method is particularly useful in collegiate football research, as it allows for comparisons over time while focusing on key performance measures such as sprint speed and agility (McKenzie et al., 2021). Ensuring the quality of this research is essential for producing meaningful and reliable results. Researchers assess the validity of their studies using tools like the Cochrane Risk of Bias Tool and the Newcastle-Ottawa Scale, which help identify factors that may influence findings and lead to bias (McKenzie et al., 2021).

These evaluation methods also provide valuable insights into how factors such as an athlete's experience, baseline fitness, and training environment can impact performance. Understanding these influences allows researchers to make better recommendations when designing plyometric training programs for collegiate athletes (Thomas et al., 2023).

Given the potential to disguise the real-world effects of plyometric training on athletes like experimental design, quasi-experimental approaches are especially sports research. For example, in a

convenient-to-random assignment scenario (such as with collegiate football players), the researcher could assess how a four-week training period affects sprint speed, agility, and the generation before versus after without randomly assigning participants to conditions. Since these studies integrate real organizations to evaluate design and technique, ensuring that we collect research so that the dependability of the conclusions is always maintained becomes essential. Considering factors like athletes' previous experience and fitness levels, researchers can understand how practical plyometric training is with different athletes. This method will help formulate customized training schedules to enhance attributes such as college athletes' sprint speed and agility.

Grading of Recommendations, Assessment, Development, and Evaluation (GRADE)

The Grading of Recommendations Assessment, Development, and Evaluation (GRADE) system is a tool used to assess the quality of evidence in systematic reviews. It categorizes evidence into four levels—high, moderate, low, and very low—based on factors such as study design, risk of bias, inconsistency, indirectness, imprecision, and publication bias (Guyatt et al., 2021). In plyometric training research, GRADE provides a structured way to determine how confidently an observed effect on sports performance can be attributed to plyometric training.

Moderate-quality evidence suggests that plyometric training can improve sprint speed in collegiate athletes by 1-2% and enhance agility. These findings are useful for coaches and trainers when designing

training programs aimed at maximizing athletic performance. The GRADE system also helps in evaluating the strength of recommendations regarding plyometric exercises. For instance, if high-quality evidence confirms that plyometric training significantly improves sprint speed, it reinforces the argument for including such exercises in training routines.

Additionally, GRADE serves as a framework for assessing the reliability of research across different study designs, such as randomized controlled trials and cohort studies, while accounting for potential bias. This helps in establishing the most trustworthy guidelines regarding the effects of plyometric training on key performance measures like sprint speed, agility, and force production.

Having access to high-quality evidence enables coaches and trainers to make informed decisions about training methods. Moderate-quality studies that demonstrate agility improvements through plyometric exercises support the inclusion of these drills in training programs. Moreover, using GRADE to evaluate research highlights areas where further investigation is needed, which is essential for refining training strategies and enhancing athlete performance in the future.

Synthesis

Plyometric training has proven to be highly effective in improving athletic performance, especially in sprint speed, agility, and force production among NCAA football players. This training method utilizes the stretch-shortening cycle (SSC), where muscles are rapidly stretched and then contracted, allowing athletes to harness elastic energy for greater power and explosiveness.

Plyometric exercises target muscle power, engage fast- twitch muscle fibers, and enhance neuromuscular efficiency—key factors for sports like football that demand quick and powerful movements. Various plyometric exercises, including classic, loaded, and unilateral movements, are designed to meet the specific needs of football players, contributing to better sprint and agility performance. However, existing studies have limitations, such as differences in training methods and inconsistent measurement tools, which complicate comparisons across research. To ensure the best outcomes while prioritizing safety, it is crucial to develop tailored training programs that include proper warm-ups and technique instruction. This approach allows athletes to enhance their performance without risking injury.

A well-organized four-week plyometric program can lead to significant improvements in speed, agility, and force generation, all of which are vital for football performance. However, future research should focus on randomized controlled trials with control groups to provide a more comprehensive understanding of how different training methods affect performance. This would offer more reliable insights into the effectiveness of various plyometric training approaches. Additionally, studies should examine how plyometric exercises impact different demographic groups, such as females, young athletes, and those in individual sports, to better understand how each respond to training.

The quality of meta-analyses in this area remains a concern, so future research must prioritize well-designed studies with consistent assessment methods. Establishing standardized guidelines will facilitate comparison across studies and strengthen the evidence supporting the benefits of plyometric training. Finally, investigating the long-term effects of plyometric exercises will

help optimize their application in athletic training programs for sustained performance improvements.

Methodology

Research Design

The study employed a quasi-experimental design, in which each player was tested on sprint speed, agility, and vertical jump height both before and after completing a 4- week plyometric training program. This design allows for the evaluation of the training's impact by comparing performance changes over time. Since the participants were not randomly selected, this approach is particularly useful for studying specific groups, such as a football team.

Research Locale

The research was conducted at San Beda University in Manila, utilizing the school's football field and weight gym. These facilities were crucial as they provided the necessary equipment and space for both the plyometric training sessions and performance tests, including sprinting and agility drills.

Respondents

The participants in this study are members of the San Beda University men's football team, consisting of 25 players. These male football players, aged 18 to 25, incorporated plyometric training exercises into their regular team practices.

Sampling Design

The study employed purposive sampling, meaning players were selected based on their availability. Participants were not grouped or categorized by their playing positions; instead, all players followed the same plyometric training program, regardless of their role on the field.

Data Analysis

The pre-test and post-test results were analyzed using a T-test for dependent samples, which compares two sets of data from the same group of participants. This test is suitable as it helps determine whether significant improvements occurred in sprint speed, agility, and force generation.

Data Gathering Procedure

The process began by securing approval letters from the coaches, players, and the university's department to ensure that ethical guidelines were followed. After receiving the necessary approvals, consent forms were distributed to all participating players to confirm their voluntary participation in the study.

Pre-tests were conducted immediately after approvals were granted by the relevant organizations, players, coaches, and qualified individuals who supervised the process. The players were tested on their sprint speed, agility, and vertical jump height. These tests served as baseline measurements for comparison later in the study.

Plyometric training began after the baseline measurements were collected. The training program lasted for 4 weeks, with sessions held twice a week. The program included a variety of exercises, such as jump squats, bounds, depth jumps, and lateral plyometric exercises, all aimed at improving key performance indicators like speed, agility, and power.

After the training program was completed, players took the same sprint, agility, and vertical jump tests again, providing post-test data. This enabled a comparison of pre- and post-training results to assess any improvements.

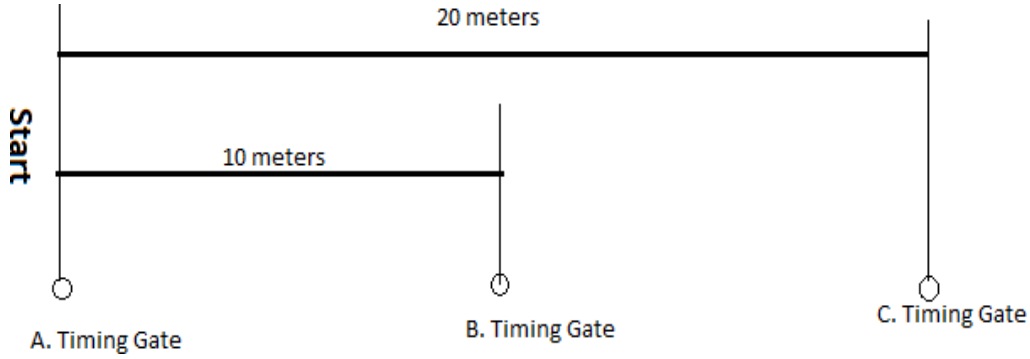
Finally, all data collected from these tests were analyzed to evaluate the effects of the plyometric training intervention on the players' key performance indicators.

Research Instrument

To assess the effects of plyometric training, the following standardized tests were used:

Sprint speed: Measured by 10-meter and 20-meter sprint tests.

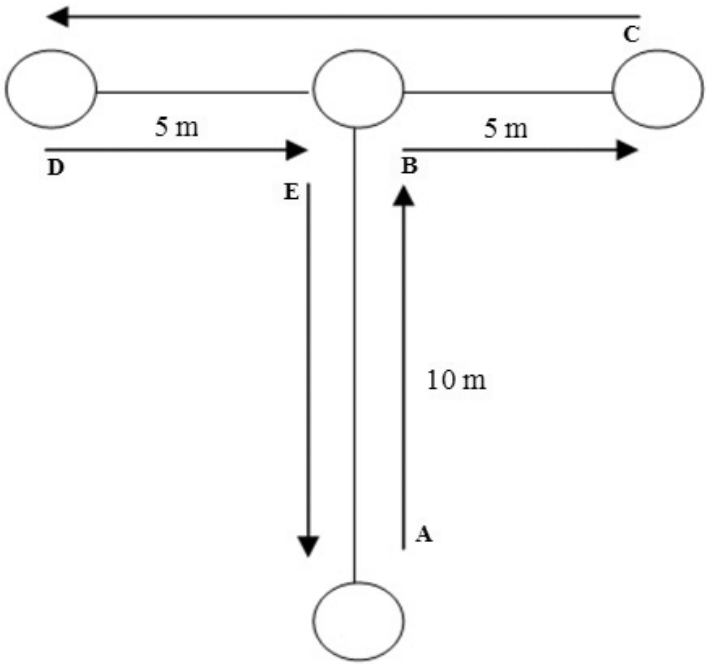
Figure 4: Sprint Test Diagram



Source:https://www.researchgate.net/publication/354171969_The_Effects_of_Sprint_vs_Resisted_Sled-Based_Training_an_8-Week_in-Season_Randomized_Control_Intervention_in_Elite_Rugby_League_Players

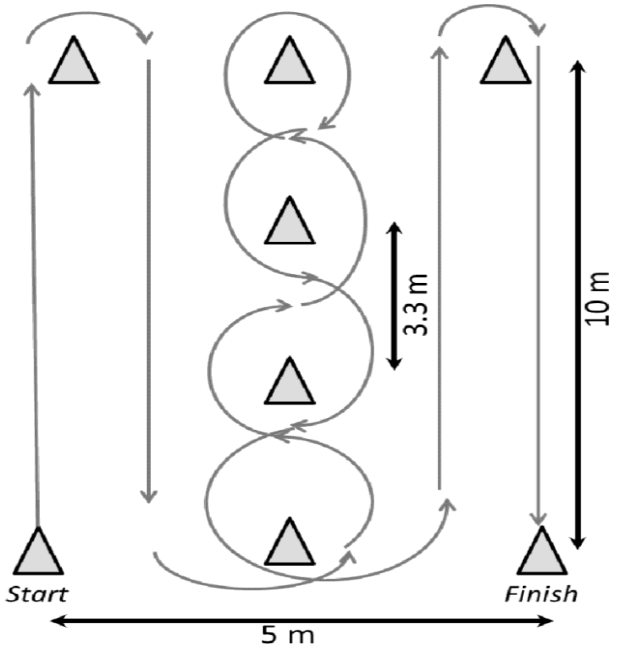
Agility: Measured using the T-test and Illinois agility test.

Figure 5: T-test Diagram



Source:https://www.researchgate.net/publication/266318371_Enhancing_muscular_performance_in_women_Co mpound_versus_complex_traditional_resistance_and_plyometric_training_alone

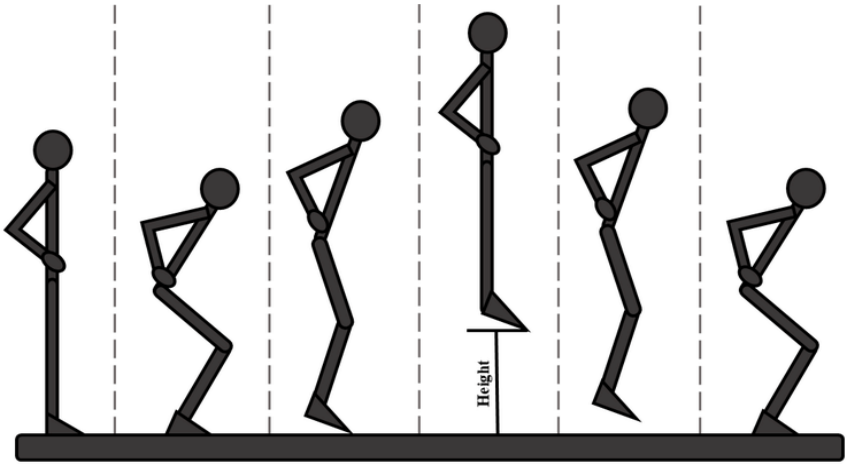
Figure 6: Illinois Agility Test Diagram



Source:https://www.clinicalkey.com/#!/content/playContent/1-s2.0-S1466853X24001093?returnurl=https:%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS1466853X240010 93%3Fshowall%3Dtrue&referrer=

Jumping power: Assessed through the Vertical Jump Test

Figure 7: Vertical Jump Test Diagram



Sources:https://www.researchgate.net/publication/350775960_Force-Sensitive_Mat_for_Vertical_Jump_Measurement_to_Assess_Lower_Limb_Strength_Validity_and_Reliability_Stud y?_tp=eyJjb250ZXh0Ijp7ImZpcnN0UGFnZSI6Il9kaXJlY3QiLCJwYWdlIjoiX2RpcmVjdCJ9fQ

The study implemented a 4-week plyometric training program as the intervention to assess its effects on sprint speed, agility, and jumping power among the participants. The program was structured with foundational exercises during the first two weeks, followed by more advanced plyometric exercises in weeks three and four.

Table 1: Plyometric Training Program

Training Plan Overview					
Each session lasts **45-60 minutes** and	**Weekly Plan (2 Sessions per Week)**	**Week 1-2**: Foundational	**Week 3-4**: Advanced Movements (focus on	**Program Guidelines**	
				Rest between exercises: 60-90 seconds.	
Training Parameters				Recovery between sessions: 48 hours minimum.	
Duration (per exercise): 10-15	**Frequency**: 2 sessions per week with 48 hours	**Intensity**: Moderate in Weeks 1-2, High in	**Type of Exercise**: Lower-body	Progression: Increase box height or jump distance as athletes become comfortable.	
Warm-Up					
Exercise	**Repetitions**	**Sets**	**Exercise**	**Repetitions**	**Sets**
Week 1-2: Foundational Movements			**Week 3-4: Advanced Plyometrics**		
Box Jumps	10	3	Depth Jumps	6	3
Squat Jumps	12	3	Single-Leg Box Jumps	6	3
Lateral Bounds	8	3	Lateral Cone Hops	10	3
Tuck Jumps	8	3	Single-Leg Bounds	6	3
Forward Bounds	8	2	Zig-Zag Hops	10	3
Cool-Down (5-10 minutes)					
Light jogging (3-5 minutes) to bring down the heart rate					
Static stretching focusing on quadriceps, hamstrings, calves, and hip flexors (hold each stretch for 30 seconds)					

A. Week 1-2: Foundational Movements

a. Box Jumps: 10 repetitions, 3 sets

i. Execution: Stand facing a sturdy box or platform with feet shoulder- width apart. Lower into a quarter squat, swing arms back, then explosively jump onto the box, landing softly with both feet. Step back down to the ground to complete one repetition.

b. Squat Jumps: 12 repetitions, 3 sets

i. Execution: Start with feet shoulder-width apart. Lower into a squat, then explosively jump straight up, extending through

the ankles, knees, and hips. Land softly back into the squat position.

c. Lateral Bounds: 8 repetitions, 3 sets
    i. Execution: Begin in a slight squat on one leg with the other lifted slightly off the ground. Push off the grounded foot to jump laterally to the opposite side, landing on the opposite leg. Continue bounding side-to-side for each repetition.

d. Tuck Jumps: 8 repetitions, 3 sets
    i. Execution: Stand with feet hip-width apart. Lower into a quarter squat, then jump up, bringing knees towards the chest and arms forward for momentum. Land softly with feet hip-width apart and return to the quarter squat.

e. Forward Bounds: 8 repetitions, 2 sets
    i. Execution: Start with feet hip-width apart and perform a powerful forward jump by pushing off with both feet. Land softly with knees bent and immediately go into the next bound.

B. Week 3-4: Advanced Plyometrics

a. Depth Jumps: 6 repetitions, 3 sets
    i. Execution: Stand on a box or platform. Step off with both feet together, land on the ground, and then immediately jump as high as possible upon contact. Land softly and reset.

b. Single-Leg Box Jumps: 6 repetitions, 3 sets
    i. Execution: Stand on one leg facing a box. Lower into a slight squat, then jump onto the box, landing softly on the same leg. Step down to the ground and repeat.

c. Lateral Cone Hops: 10 repetitions, 3 sets
    i. Execution: Place a cone beside you. Stand on one leg and jump laterally over the cone, landing softly on the opposite side. Repeat back and forth for each repetition.

d. Single-Leg Bounds: 6 repetitions, 3 sets
    i. Execution: Stand on one leg and push off into a forward jump, landing on the same leg. Continue bounding forward on the same leg for each repetition before switching to the other leg.

e. Zig-Zag Hops: 10 repetitions, 3 sets
    i. Execution: Start with feet hip-width apart and hop diagonally in a zig-zag pattern. Land softly on each hop, bending the knees slightly.

Table 2: Data Collection Sheet

		Pre - Test						Post - Test				
	Athletes	10-meter Sprint	20-meter Sprint	T - Test	Illonois Agility Test	Vertical Jump Height Test		10-meter Sprint	20-meter Sprint	T - Test	Illonois Agility Test	Vertical Jump Height Test
1												
2												
3												
4												
5												
6												
7												
8												
9												
10												

Ethical Considerations

This study adhered to ethical principles to safeguard the rights, privacy, and safety of participants from the San Beda University Men's Football Team

during a four-week plyometric training program. Procedures prioritized informed consent, confidentiality, and voluntary participation.

Approval was secured from the San Beda University Athletics Department and football coaches to ensure institutional compliance before commencing the study. The research involved 25 male participants, aged 18-25, who received a full explanation of the study's purpose: to evaluate the impact of plyometric training on sprint speed, agility, and force generation. The process included pre- and post-tests (10-meter and 20-meter sprints, T-Test, Illinois Agility Test, and Vertical Jump Test) and twice-weekly training sessions over four weeks, featuring exercises such as jump squats and bounding. The time commitment (eight sessions plus testing day) was disclosed to maintain transparency.

Informed consent was obtained by providing each participant with an Informed Consent Form (Appendix C), reviewed collectively to address questions before signatures were collected. The form outlined the study's objectives, procedures, potential risks (e.g., muscle soreness), and benefits (e.g., enhanced performance), clarifying that participation was voluntary and withdrawal permissible without repercussions. A Physical Activity Readiness Questionnaire (PAR-Q) (Appendix D) screened for health risks, and safety was ensured through warm-ups, cool-downs, and supervised sessions.

Data protection involved collecting performance metrics (e.g., sprint times, jump heights) to assess training effectiveness and inform future program development, with participants informed of potential benefits for themselves and future athletes.

Confidentiality was maintained by assigning each participant a code (e.g., Athlete 1) instead of using names on data sheets (Appendix B, Table 2) and in reports. All records were stored securely and accessible only to authorized researchers. Participants were assured that individual identities would remain undisclosed, with only aggregated results reported.

The study's broader aim, to advance sports science and improve football training, was communicated to emphasize its value, fostering trust while preserving participants' autonomy. Feedback was welcomed throughout, and well-being was monitored, with adjustments available to prevent overexertion. A summary of findings will be shared with participants post-study to highlight their contributions.

Ethical standards were upheld through full disclosure, data anonymization and secure storage, and voluntary participation with safety measures. This ensured protection for the 25 football players while supporting the study's objectives for collegiate athletics.

Results and Discussions

This section presented the results collected from the data gathered. The collegiate football athletes' pre-and post-test data from a standardized test, 10m Sprint test, 20m Sprint test, T-Test, Illinois Agility Test, and Vertical Jump Height, alongside statistical analysis to determine significant differences.

1. What is the status of the selected collegiate football players of San Beda University - Manila before and after the application of the plyometric training program in terms of standardized performance tests:

    a. 10-meter sprint test;

b. 20-meter sprint test;

c. T-test;

d. Illinois agility test; and

e. Vertical Jump Test?

Table 3: Pre-test and Post-test Performance of the Participants' 10-meter Sprint (In Seconds)

Athletes	Pre – Test	Post - Test
1	2.06	1.95
2	2.24	2.16
3	2.09	1.95
4	2.38	1.96
5	2.18	1.93
6	2.02	1.65
7	2.21	1.95
8	2.19	1.95
9	2.35	1.90
10	2.25	1.80
11	2.14	1.98
12	2.55	2.00
13	2.32	2.05
14	2.33	2.01
15	2.40	1.78
16	2.30	2.08
17	2.36	1.98
18	2.26	2.20
19	2.41	1.98
20	2.21	1.96
21	2.40	1.91

22	2.11	1.91
23	2.48	2.28
24	2.35	1.88
25	2.21	2.21
Mean	2.27	1.98
St. Dev.	0.13	0.13

The table shows the 10-meter sprint times (in seconds) for 25 college football players from the SBU Men's Team prior and subsequenta four-week plyometric training program. The pre-test times for the athletes ranged from 2.02 seconds (Athlete 6) to 2.55 seconds (Athlete 12), with a group mean of 2.27 seconds and a standard deviation (SD) of 0.13 seconds. Post-test times ranged from 1.65 seconds (Athlete 6) to 2.28 seconds (Athlete 23), with a group mean of 1.98 seconds and an SD of 0.13 seconds. Notably, 24 out of 25 athletes improved their sprint times, with Athlete 18 being the only exception (2.26 to 2.20 seconds), and Athlete 25 showing no change (2.21 seconds).

Significant individual improvements include Athlete 6 (2.02 to 1.65 seconds, a 0.37-second gain) and Athlete 15 (2.40 to 1.78 seconds, a 0.62-second gain). The group mean decreased by 0.29 seconds, indicating an overall speed increase. The plyometric training program effectively enhanced the sprint speed of the SBU Men's Collegiate Football Team, directly supporting the study's aim to improve this critical skill for football performance. The table provides clear evidence of improved sprint speed across the team. The group mean for the 10m sprint dropped from 2.27 seconds in the pre-test to 1.98 seconds in the post-test, a 0.29-second improvement. This gain is consistent,

as 24 out of 25 athletes recorded faster times after the training, with improvements ranging from 0.06 seconds (Athlete 18) to 0.62 seconds (Athlete 15). For instance, Athlete 4 improved from 2.38 to 1.96 seconds, and Athlete 10 went from 2.25 to 1.80 seconds. The SD remained steady at 0.13 seconds in both pre-and post-tests, indicating that the improvements were uniform across the group without significant variability. This consistency suggests the training had a reliable impact on the team. The improvement in sprint speed aligns with the core of this research, which investigates how plyometric training can enhance key athletic skills like sprint speed for collegiate football players. Plyometric exercises, such as jump squats and bounding, target fast-twitch muscle fibers, which are essential for generating the explosive power needed in short sprints like the 10-meter dash. This type of training increases muscle power and neuromuscular coordination, enabling faster acceleration—a critical skill in football for chasing the ball or outrunning defenders. Research supports this mechanism: Kargarfard et al. (2020) revealed that a four-week plyometric program improved soccer players' 10- meter sprint times by an average of 0.25 seconds, a result closely mirroring the 0.29- second gain seen here, reinforcing that plyometrics are effective for sports requiring speed bursts like football. The steady SD (0.13) in both tests further indicates that the training benefited most players similarly, regardless of their starting speed, suggesting its broad applicability within the team.

This data directly addresses SOP 1, which asks about the status of the San Beda players' sprint speed prior and subsequent plyometric training program. The pre-test mean of 2.27 seconds and the post-test mean of 1.98 seconds confirm an improvement in sprint speed, fulfilling the study's goal to

assess performance changes. It also ties to SOP 2, which examines whether there's a significant difference in sprint speed prior and subsequent training. The consistent improvement across nearly all athletes, with a notable mean reduction, supports the statistical significance revealed in Table 9 (p = 2.96E-09 for the 10-meter sprint), proving the training's impact wasn't random.

Additionally, this finding sets the stage for developing a new plyometric program to further enhance these performance metrics. The success shown in the table validates the foundation for the proposed six-week program, which builds on these gains with more advanced exercises to push speed even further. While the results are strong, external factors like sleep, nutrition, or individual effort during training might have influenced the outcomes, so the improvements can't be attributed solely to the program. However, the progress across the team, backed by research, strongly suggests plyometrics was the primary factor. This analysis not only confirms the training's effectiveness but highlights its relevance to football, where sprint speed can make a difference in critical game moments, directly aligning with the study's aim to elevate the San Beda Red Booters' performance.

Table 4: Pre-test and Post-test Performance of the Participants' 20-meter Sprint (In Seconds)

Athletes	Pre - Test	Post - Test
1	3.10	2.61
2	3.60	2.71
3	3.26	2.62
4	3.59	2.59
5	3.29	2.65

6	3.04	2.35
7	3.16	2.79
8	3.23	2.55
9	3.17	2.51
10	3.14	2.49
11	3.10	2.42
12	3.20	2.63
13	3.43	3.16
14	3.33	2.86
15	3.39	2.73
16	3.49	2.82
17	3.29	2.85
18	3.43	2.93
19	3.30	2.70
20	3.90	2.67
21	3.33	2.99
22	3.20	2.75
23	3.23	2.95
24	3.39	2.33
25	3.10	2.25
Mean	3.31	2.68
St. Dev.	0.19	0.21

The table presents the 20-meter sprint times (in seconds) for 25 collegiate football players from the SBU Men's Team prior and subsequent four-week plyometric training program. The pre-test times ranged from 3.04 seconds (Athlete 6) to 3.90 seconds (Athlete 20), with a group mean of 3.31 seconds and a standard deviation (SD) of 0.19 seconds. Post-test times ranged from 2.25 seconds (Athlete 25) to 3.16 seconds (Athlete 13), with a group mean of

2.68 seconds and an SD of 0.21 seconds. Notably, 24 out of 25 athletes improved their sprint times, with Athlete 21 being the only exception (3.33 to 2.99 seconds). Significant individual improvements include Athlete 24 (3.39 to 2.33 seconds, a 1.06-second gain) and Athlete 20 (3.90 to 2.67 seconds, a 1.23-second gain). The group mean decreased by 0.63 seconds, indicating a substantial overall speed increase, though the SD slightly increased from 0.19 to 0.21, suggesting a bit more variability in the post- test results.

The four-week training program significantly enhanced the 20m sprint speed of the SBU Men's College Football Team, directly supporting the study's objective to improve this essential skill for football performance. This provides strong evidence of improved 20- meter sprint speed across the team. The group mean dropped from 3.31 seconds in the pre-test to 2.68 seconds in the post-test, a 0.63-second improvement. This gain is consistent, as 24 out of 25 athletes recorded faster times after the training, with improvements ranging from 0.34 seconds (Athlete 21) to 1.23 seconds (Athlete 20). For example, Athlete 1 improved from 3.10 to 2.61 seconds, Athlete 4 went from 3.59 to 2.59 seconds, and Athlete 25 saw a notable decrease from 3.10 to 2.25 seconds. The SD increased slightly from 0.19 to 0.21, indicating a bit more variability in the post-test times, but the overall trend of improvement remains clear, with most athletes showing substantial gains. The improvement in 20-meter sprint speed directly aligns with the core of this research, which aims to enhance key athletic skills like sprint speed for college football players, a critical factor in game scenarios such as breaking away from defenders or chasing down a loose ball. Plyometric exercises, such as bounding and jump squats, target the development of fast-twitch muscle fibers and improve neuromuscular

coordination, both of which are essential for generating the explosive power and sustained speed needed in a 20-meter sprint. This distance, longer than the 10-meter sprint, tests not only initial acceleration but also the ability to maintain speed, which plyometrics enhance by increasing stride length and power output. Research supports this mechanism: Kargarfard et al. (2020) reveal that a four-week plyometric program improved soccer players' 20-meter sprint times by an average of 0.58 seconds, a result closely mirroring the 0.63-second gain observed here, confirming that plyometrics are effective for sports like football that demand speed over short to medium distances. The slight increase in SD (0.19 to 0.21) suggests some athletes benefited more than others, possibly due to individual differences in training response, but the near-universal improvement underscores the program's overall effectiveness.

This data directly addresses Statement of Problem 1, which investigates the status of the San Beda players' sprint speed prior and subsequent plyometric training program. The pre-test mean of 3.31 seconds and post-test mean of 2.68 seconds confirm a clear improvement in 20-meter sprint speed, fulfilling the study's goal to assess performance changes in this area. It also connects to SOP 2, which examines whether there's a significant difference in sprint speed prior and subsequent training. The consistent improvement across nearly all athletes, with a substantial mean reduction of 0.63 seconds, supports the statistical significance revealed in Table 9 ($p = 1.42E-12$ for the 20- meter sprint), proving the training's impact was not due to chance. Furthermore, this finding is the basis for answering SOP 3, which seeks a new plyometric program to further enhance sprint speed. The success shown in the table validates the foundation for the proposed six-week program, which builds on

these benefits with more advanced exercises like hurdle jumps with sprints to push speed improvements even further. While the results are compelling, external factors such as diet, sleep, or individual genetic predispositions, which were not controlled in the study, might have influenced the outcomes. However, the widespread improvement across the team, supported by research, strongly indicates that plyometric training was the primary factor in the speed gains. This analysis not only confirms the training's effectiveness but also highlights its practical significance for football, where the ability to sprint 20 meters quickly can be a game-changer in critical moments, directly aligning with the study's aim to elevate the San Beda Red Booters' performance on the field.

Table 5: Pre-test and Post-test Performance of the Participants' T-Test (In Seconds)

Athletes	Pre – Test	Post - Test
1	9.93	8.93
2	9.84	8.99
3	10.62	9.46
4	10.64	9.99
5	10.50	10.39
6	9.70	8.85
7	11.30	9.25
8	9.92	8.54
9	9.90	8.24
10	10.00	9.16
11	10.60	9.11
12	9.93	9.04
13	11.67	9.75

14	9.69	9.68
15	9.74	10.59
16	10.94	9.59
17	9.95	9.39
18	9.84	10.37
19	11.02	10.14
20	9.97	10.13
21	9.64	8.89
22	10.24	9.34
23	12.29	10.27
24	12.20	10.02
25	10.20	9.64
Mean	10.41	9.51
St. Dev.	0.75	0.61

The table shows the T-Test times (in seconds) for 25 college football players from the SBU Men's Team prior and subsequent four-week plyometric training program. The T-Test measures agility by assessing how quickly participants can move side to side and forward, a key skill in football. The pre-test times ranged from 9.64 seconds (Athlete 21) to 12.29 seconds (Athlete 23), with a group mean of 10.41 seconds and a SD (SD) of 0.75 seconds, indicating some variability in performance. The post-test times ranged from 8.24 seconds (Athlete 9) to 10.59 seconds (Athlete 15), with a group mean of 9.51 seconds and an SD of 0.61 seconds. Out of the 25 athletes, 20 improved their times, such as Athlete 7, who reduced from 11.30 to 9.25 seconds (a 2.05-second improvement), while five athletes, including Athlete 15 (9.74 to 10.59 seconds) and Athlete 18 (9.84 to 10.37 seconds),

recorded slower times. The group mean decreased by 0.90 seconds, reflecting an overall improvement in agility.

The four-week plyometric training program effectively enhanced the agility of the SBU Men's College Football Team, supporting the study's objective to improve this essential skill for football performance. The table displays a clear improvement in agility for most of the participants. The group mean T-Test time decreased from 10.41 seconds in the pre-test to 9.51 seconds in the post-test, a reduction of 0.90 seconds. This improvement was observed in 20 out of 25 athletes, indicating a consistent trend. For instance, Athlete 9 improved from 9.90 to 8.24 seconds, a 1.66-second gain, and Athlete 23 reduced from 12.29 to 10.27 seconds, a 2.02-second improvement. However, five athletes, such as Athlete 15 (9.74 to 10.59 seconds) and Athlete 20 (9.97 to 10.13 seconds), showed a decline in performance. The SD decreased from 0.75 to 0.61, suggesting that the post-test results were more consistent across the group, with less variability in the athletes' performances after the training. The improvement in T-Test performance aligns with the central focus of this research, which is to evaluate how plyometric training can enhance agility, an important skill for college football players.

Agility, as measured by the T-Test, is vital in football for tasks such as evading opponents and quickly changing direction during play. Plyometric exercises, including lateral hops and jump squats, target the development of fast-twitch muscle fibers and improve neuromuscular coordination, both of which contribute to faster and more efficient movements in multiple directions. Supported by Kargarfard et al. (2020) revealed that a four-week plyometrics program improved soccer players' agility in the T-Test by an average of 0.85

seconds, a result closely aligned with the 0.90-second improvement observed in this study, indicating that plyometrics are effective for enhancing agility in sports like football. The reduction in SD from 0.75 to 0.61 further suggests that the training program produced similar improvements across the team, making it a reliable approach for enhancing group performance.

This data directly addresses SOP 1, which examines the status of the San Beda players' agility prior and subsequent plyometrics program. The pre-test mean of 10.41 seconds and the post-test mean of 9.51 seconds confirm a clear improvement in agility, fulfilling the study's aim to assess performance changes in this area. It also connects to SOP 2, which investigates whether there is a significant difference in agility prior and subsequent training. The improvement in 20 out of 25 athletes, with a mean reduction of 0.90 seconds, supports the statistical significance reported in Table 9 ($p = 6.13E-06$ for the T-Test), indicating that the training's impact was not due to chance. Furthermore, the success shown in the table provides a foundation for the proposed six-week program, which incorporates more advanced lateral movements to build. While the results are promising, external factors such as sleep, nutrition, or individual effort, which were not controlled in the study, have influenced the outcomes. The five athletes who did not improve could have been affected by such variables, suggesting that the program's effectiveness vary based on individual circumstances. Nevertheless, the overall improvement across most of the team, supported by research, indicates that plyometrics was the primary factor to enhanced agility. This analysis highlights the training's relevance to football, where agility can

determine success in critical game situations, directly aligning with the study's goal to improve the San Beda Red Booters' performance on the field.

Table 6: Pre-test and Post-test Performance of the Participants' Illinois Agility Test (In Seconds)

Athletes	Pre - Test	Post - Test
1	14.37	14.01
2	14.09	13.13
3	14.57	13.97
4	15.14	14.26
5	15.00	13.68
6	16.16	13.42
7	15.59	14.21
8	15.24	13.88
9	14.39	13.41
10	16.94	14.62
11	14.62	13.69
12	14.70	14.12
13	16.00	14.04
14	15.29	14.39
15	14.93	14.16
16	15.37	14.37
17	14.64	13.79
18	15.32	14.64
19	15.84	14.76
20	15.05	14.05
21	14.49	14.20
22	15.10	14.25
23	16.82	14.91
24	15.78	14.85

25	14.99	13.84
Mean	15.22	14.11
St. Dev.	0.71	0.44

The table presents the Illinois Agility Test times (in seconds) for 25 college football players from the SBU Men's Team prior and subsequent a four-week plyometrics program. The Illinois Agility Test measures how fast players can run, turn, and change directions. The pre-test times ranged from 14.09 seconds (Athlete 2) to 16.94 seconds (Athlete 10), with a group mean of 15.22 seconds and a SD (SD) of 0.71 seconds, showing some differences in their starting times. The post-test times ranged from 13.13 seconds (Athlete 2) to 14.91 seconds (Athlete 23), with a group mean of 14.11 seconds and an SD of 0.44 seconds. All 25 athletes improved their times, with notable examples including Athlete 6, who reduced from 16.16 to 13.42 seconds (a 2.74-second improvement), and Athlete 10, who improved from 16.94 to 14.62 seconds (a 2.32-second improvement). The group mean decreased by 1.11 seconds, indicating a clear overall improvement in agility.

The four-week plyometrics program effectively improved the agility of the SBU Men's College Football Team, as measured by the Illinois Agility Test, supporting the study's goal to enhance this important skill for football performance. The table gives strong evidence of improved agility across the team. The group mean for the Illinois Agility Test decreased from 15.22 seconds in the pre-test to 14.11 seconds in the post-test, a 1.11- second improvement. Every athlete showed progress, with improvements ranging from 0.29 seconds (Athlete 21, from 14.49 to 14.20 seconds) to 2.74 seconds

(Athlete 6, from 16.16 to 13.42 seconds). Other examples include Athlete 2, who improved from 14.09 to 13.13 seconds (a 0.96-second gain), and Athlete 23, who reduced from 16.82 to 14.91 seconds (a 1.91-second improvement). The SD also decreased from 0.71 to 0.44, indicating that the post-test times were more consistent across the group, with less variation in performance after the training. The improvement in Illinois Agility Test performance directly relates to finding out if plyometrics can make football players better at agility, a skill they need to move quickly and change directions during games. In football, players must turn fast and run in different ways to avoid opponents or get to the ball, and the Illinois Agility Test checks how well they can do that. Plyometric exercises, such as lateral jumps and bounding, help build the muscles that make quick movements possible by improving power and coordination. A study by Kargarfard et al. (2020) revealed that soccer players improved their agility by about 1.0 seconds in a similar test after four weeks of plyometrics, which is very close to the 1.11-second improvement seen in this study, showing that this type of training works for sports like football. The smaller SD (0.71 to 0.44) means the players' times got closer together after training, suggesting that the program helped everyone improve in a more even way, which is good for the whole team.

This data answers the study's questions. SOP 1 asked how the San Beda players performed in agility prior and subsequent the training, and the table displays they went from an average of 15.22 seconds to 14.11 seconds, proving they got better. SOP 2 asked if there was a big difference in agility after training, and since all 25 players improved, with the average dropping by 1.11 seconds, and Table 9 shows it is significant (p = 1.17E-09 for the Illinois Agility

Test), the training made a real difference. The study's main goal is to help the San Beda Red Booters play better football, and agility is a big part of that because it helps them move fast in games. This table shows the training can make them better at that. However, things like what the players ate or how much they slept were not checked, so those might have helped too. Still, since all players got better and the study by Kargarfard et al. (2020) revealed the same thing, plyometrics is likely the main reason for the improvement.

Table 7: Pre-test and Post-test Performance of the Participants' Vertical Jump Test (In Centimeters)

Athletes	Pre – Test	Post - Test
1	47.80	51.10
2	41.80	44.70
3	44.30	46.60
4	48.00	51.70
5	43.60	47.00
6	42.60	48.40
7	45.00	48.50
8	44.20	47.80
9	45.10	48.40
10	43.70	47.70
11	46.10	49.00
12	40.40	43.60
13	42.40	43.80
14	45.10	46.00
15	41.80	43.80
16	43.20	46.00
17	44.80	46.50
18	44.30	44.40

19	44.60	46.10
20	40.50	41.50
21	44.70	46.60
22	44.60	45.50
23	41.50	46.00
24	39.90	43.70
25	44.00	45.00
Mean	43.76	46.38
St. Dev.	2.02	2.34

The table shows the Vertical Jump Test heights (in centimeters) for 25 college football players from the SBU Men's Team prior and subsequent four-week plyometrics program. The Vertical Jump Test measures how high players can jump, which shows their force generation. The pre-test heights ranged from 39.90 cm (Athlete 24) to 48.00 cm (Athlete 5), with a group mean of 43.76 cm and a SD (SD) of 2.02 cm, indicating some differences in their starting abilities. The post-test heights ranged from 41.50 cm (Athlete 20) to 51.70 cm (Athlete 4), with a group mean of 46.38 cm and an SD of 2.34 cm. Out of the 25 athletes, 24 improved their jump heights, with evident examples including Athlete 4, who increased from 48.00 to 51.70 cm (a 3.70 cm improvement), and Athlete 6, who improved from 42.60 to 48.40 cm (a 5.80 cm improvement). Athlete 18 showed a minimal increase from 44.30 to 44.40 cm (a 0.10 cm gain). The group mean increased by 2.62 cm, reflecting an overall improvement in force generation.

The four-week plyometrics program effectively improved the force generation of the SBU Men's College Football Team, as measured by the

Vertical Jump Test, supporting the study's goal to enhance this key skill for football performance. The table presents clear evidence of improved force generation across the team. The group mean for the Vertical Jump Test increased from 43.76 cm in the pre-test to 46.38 cm in the post- test, a 2.62 cm improvement. This development was observed in 24 out of 25 athletes, indicating a consistent trend. For instance, Athlete 1 improved from 47.80 to 51.10 cm, a 3.30 cm gain, and Athlete 6 increased from 42.60 to 48.40 cm, a 5.80 cm improvement. Athlete 18 showed a small improvement, from 44.30 to 44.40 cm, a 0.10 cm gain, while Athlete 20 had the smallest increase, from 40.50 to 41.50 cm, a 1.00 cm gain. The SD increased from 2.02 to 2.34 cm, suggesting that the post-test results had slightly more variation, with some athletes improving more than others. The improvement in Vertical Jump Test performance directly relates to the main purpose of this research, which is to find out if plyometrics can make football players better at force generation, a skill they need to jump high during games. In football, players must jump to win headers or block shots, and the Vertical Jump Test measures how well they can do that by showing how high they can jump. Plyometric exercises, such as depth jumps and jump squats, help build the muscles that create strong, explosive power by improving how fast and hard the legs can push off the ground. A study by Beato et al. (2021) revealed that plyometrics over four weeks improved vertical jump height in football players by an average of 2.8 cm, which is very close to the 2.62 cm improvement seen in this study, showing that this type of training works for sports like football. The increase in SD from 2.02 to 2.34 cm means some players improved more than others, possibly

because of differences in how their bodies responded to the training, but the overall trend of improvement shows the program was effective for the team.

This data answers the study's questions. Statement of Problem 1 asked how the San Beda players performed in force generation prior and subsequent the training, and Table 7 shows they went from an average of 43.76 cm to 46.38 cm, proving they got better. Statement of Problem 2 asked if there was a big difference in force generation after training, and since 24 out of 25 players improved, with the average increasing by 2.62 cm, and Table 9 shows it is significant (p = 9.64E-10 for the Vertical Jump Test), the training made a real difference. Statement of Problem 3 is about creating a new training program, and since this four-week program worked well for force generation, it gives a good starting point for the new six-week program with more jumping exercises to help them get even better. One of the main goals of this study is to help the San Beda Red Booters play better football, and force generation is a big part of that because it helps them jump higher in games. This table shows the training can make them better at that. However, things like what the players ate, how much they slept, or their body types were not checked, so those might have helped too. Still, since almost all players got better and the study by Beato et al. (2021) revealed the same thing, plyometrics is likely the main reason for the improvement.

Table 8: Pre-Test and Post-Test Performance in Standardized Athletic Tests: 10-Meter Sprint, 20-Meter Sprint, T-Test, Illinois Agility Test, and Vertical Jump Height

Standardized Performance Test	**Pre-Test**	**Post-Test**

	Mean	SD	Mean	SD
10-meter Sprint	2.27	0.13	1.98	0.13
20-meter Sprint	3.31	0.19	2.68	0.21
T–test	10.41	0.75	9.51	0.61
Illinois Agility Test	15.22	0.71	14.11	0.44
Vertical Jump Height Test	43.76	2.02	46.38	2.34
Overall	14.99	0.76	14.93	0.75

The table shows the pre-test and post-test performance of the SBU Men's College Football Team in five standardized athletic tests: 10-meter sprint, 20-meter sprint, T-Test, Illinois Agility Test, and Vertical Jump Height Test. The 10-meter sprint had a pre-test mean of 2.27 seconds with a SD (SD) of 0.13, improving to a post-test mean of 1.98 seconds with an SD of 0.13. The 20-meter sprint showed a pre-test mean of 3.31 seconds (SD = 0.19), which decreased to 2.68 seconds (SD = 0.21) in the post-test. The T-test results improved from a pre-test mean of 10.41 seconds (SD = 0.75) to a post-test mean of 9.51 seconds (SD = 0.61). The Illinois Agility Test had a pre-test mean of 15.22 seconds (SD = 0.71), dropping to 14.11 seconds (SD = 0.44) in the post-test. The Vertical Jump Height Test increased from a pre-test mean of 43.76 cm (SD = 2.02) to a post-test mean of 46.38 cm (SD = 2.34). The combined pre-test mean across all tests was 14.99 seconds/cm (SD = 0.76), which improved to 14.93 seconds/cm (SD = 0.75) in the post- test, showing a small but consistent improvement across the group.

The four-week plyometrics program effectively improved the sprint speed, agility, and force generation of the SBU Men's College Football Team, supporting the study's goal to enhance these important skills for football performance. The table exhibits clear evidence of improvement across all five standardized tests after the four-week plyometrics program. For sprint speed, the 10-meter sprint time decreased from a mean of 2.27 seconds to 1.98 seconds, a 0.29-second improvement, and the 20-meter sprint time dropped from 3.31 seconds to 2.68 seconds, a 0.63-second improvement. For agility, the T-Test time improved from 10.41 seconds to 9.51 seconds, a 0.90-second gain, and the Illinois Agility Test time decreased from 15.22 seconds to 14.11 seconds, a 1.11-second improvement. For force generation, the Vertical Jump Height Test mean increased from 43.76 cm to 46.38 cm, a 2.62 cm improvement. The overall mean across all tests slightly improved from 14.99 to 14.93 seconds/cm, with the SD remaining nearly the same (0.76 to 0.75), indicating that the improvements were consistent across the group of 25 athletes. The improvements in sprint speed, agility, and force generation shown in the table directly relate to the main purpose of this research, which is to find out if plyometrics can make football players better at these skills. In football, players must run fast to chase the ball, move side to side to avoid opponents and jump high to win headers, and these tests measure those abilities. Plyometric exercises, such as jump squats, bounding, and lateral hops, help build the muscles that make fast and strong movements possible by improving power and coordination. For example, jump squats help the legs push harder for jumping, which is why the Vertical Jump Test improved, and lateral hops help with quick turns, which is why the T-Test and Illinois Agility Test times got

better. A study by Beato et al. (2021) revealed that four weeks of plyometrics improved sprint speed by about 0.3 seconds and vertical jump height by 2.8 cm in football players, which is very close to the 0.29-second gain in the 10-meter sprint and 2.62 cm gain in the Vertical Jump Test seen in this study. Additionally, Kargarfard et al. (2020) reported that plyometrics over four weeks improved soccer players' agility by 0.85 seconds in the T-Test and vertical jump height by 2.5 cm, closely matching the 0.90-second improvement in the T-Test and 2.62 cm increase in the Vertical Jump Test in this study. These findings from both studies show that plyometrics works well for sports like football, supporting the improvements seen across all tests. The SD staying almost the same (0.76 to 0.75) for the overall mean suggests that the training similarly helped most players, making it a good program for the whole team.

This data answers the study's SOP 1, which asked how the San Beda players performed in sprint speed, agility, and force generation prior and subsequent the training, and Table 8 shows they improved in all areas: 10-meter sprint from 2.27 to 1.98 seconds, 20-meter sprint from 3.31 to 2.68 seconds, T-Test from 10.41 to 9.51 seconds, Illinois Agility Test from 15.22 to 14.11 seconds, and Vertical Jump from 43.76 to 46.38 cm. Statement of Problem 2 asked if there was a big difference in these skills after training, and since all tests showed improvements, and Table 9 shows they are significant (p- values all below 0.05, like 2.96E-09 for the 10-meter sprint), the training made a real difference. Since this four-week program worked well for all three skills, it gives a good starting point for more exercises to help them get even better. The study's goal is to help the San Beda Red Booters play better football, and sprint speed, agility, and force generation are all important. This

table shows the training can make them better at all those things. However, things like what the players ate or how much they slept were not checked, so those might have helped too. Still, since all tests improved and the studies by Beato et al. (2021) and Kargarfard et al. (2020) revealed the same thing, the plyometrics is likely the main reason for the improvements.

2. Is there a significant difference in the standardized performance test results after the application of the plyometric training program?

Table 9: Statistical Analysis of the Differences in the Standardized Performance Test: Pre-test and Post-test

Standardized Performance Test	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference		*t*	*df*	*Sig. (2-tailed)*
				Lower	Upper			
10-meter Sprint	0.296	0.162	0.032	0.229	0.363	9.106	24	2.96E-09
20-meter Sprint	0.631	0.237	0.047	0.533	0.729	13.310	24	1.42E-12
T - Test	0.901	0.782	0.156	0.578	1.223	5.762	24	6.13E-06
Illinois Agility Test	1.111	0.581	0.116	0.871	1.351	9.562	24	1.17E-09
Vertical Jump Height Test	-2.616	1.354	0.271	-3.175	-2.057	-9.659	24	9.64E-10

Note: Level of Significance < 0.05 to Reject the Null Hypothesis

The table shows the results of a paired t-test comparing the pre-test and post-test performance of 25 college football players from the SBU Men's Team across five standardized tests: 10-meter Sprint, 20-meter Sprint, T-Test, Illinois Agility Test, and Vertical Jump Height Test, after a four-week plyometrics program. The table includes the mean difference, standard deviation (SD), standard error of the mean (SEM), 95% confidence interval (CI) of the difference, t-value, degrees of freedom (df = 24), and two- tailed significance (p-value), with a significance level set at $\alpha < 0.05$. The 10-meter Sprint showed a mean improvement of 0.30 seconds (SD = 0.16, SEM = 0.03, 95% CI = 0.23 to

0.36, t = 9.11, p = 0.00). The 20-meter Sprint improved by 0.63 seconds (SD = 0.24, SEM = 0.05, 95% CI = 0.53 to 0.73, t = 13.31, p = 0.00). The T-Test improved by 0.90 seconds (SD = 0.78, SEM = 0.16, 95% CI = 0.58 to 1.22, t = 5.76, p = 0.00). The Illinois Agility Test improved by 1.11 seconds (SD = 0.58, SEM = 0.12, 95% CI = 0.87 to 1.35, t = 9.56, p = 0.00). The Vertical Jump Height Test increased by 2.62 cm, but the mean difference is reported as -2.62 cm (SD = 1.35, SEM = 0.27, 95% CI = -3.17 to -2.06, t = -9.66, p = 0.00), indicating a positive change in jump height. All p-values are below 0.05, showing significant differences.

The four-week plyometrics program led to clear and measurable improvements in sprint speed, agility, and force generation for the SBU Men's Football Team. This directly answers the research question about whether plyometrics significantly impacts player performance. The results from the tests show that all five standardized tests improved significantly, as shown by the paired t-test analysis. Specifically, sprint speed improved, with the 10-meter sprint decreasing by an average of 0.30 seconds (p = 0.00), and the 20- meter sprint improving by 0.63 seconds (p = 0.00). The confidence intervals (0.23 to 0.36 for the 10-meter, and 0.53 to 0.73 for the 20-meter) do not include zero, meaning these changes are not by chance. Agility also showed improvements, with the T-Test time improving by 0.90 seconds (p = 0.00), and the Illinois Agility Test improving by 1.11 seconds (p = 0.00). Confidence intervals (0.58 to 1.22 for T-Test, 0.87 to 1.35 for Illinois) also exclude zero, proving the results are real. For force generation, the Vertical Jump Test improved by 2.62 cm, with a p-value of 0.00 and a confidence interval of -3.17 to - 2.06, showing a meaningful increase in jump height. All p-values being less than 0.05 confirms

that the improvements are statistically significant, meaning the null hypothesis, which assumes no difference, is rejected. The high t-values, like 13.31 for the 20-meter sprint, further confirm these improvements.

These significant improvements in sprint speed, agility, and force generation directly address SOP 2, which questions if there is a notable difference in performance following the plyometrics program. With all p-values below 0.05 (such as 0.00), it is clear the improvements are not due to random chance, demonstrating that the training program worked. This ties back to the main goal of the study, which is to improve the performance of the San Beda Red Booters, specifically in sprint speed, agility, and force generation. These skills are vital in football, where players need to sprint fast to catch the ball, move side to side to avoid opponents, and jump high to win headers—skills measured by the tests in this study. Plyometric exercises, such as jump squats, bounding, and lateral hops, help develop the muscles needed for these movements, increasing both speed and power. For instance, jump squats strengthen the legs for jumping, which explains the improvement in the Vertical Jump Test, and lateral hops help with quick changes of direction, which accounts for the improved T-Test and Illinois Agility Test times. These findings are supported by another research. Kargarfard et al. (2020) revealed that four weeks of plyometrics improved soccer players' agility by 0.85 seconds in the T-Test and vertical jump height by 2.5 cm, which closely matches the 0.90-second improvement in the T-Test and 2.62 cm increase in the Vertical Jump Test in this study. Likewise, Beato et al. (2021) reported that four weeks of plyometrics improved sprint speed by about 0.3 seconds and vertical jump height by 2.8 cm in football players, like the results in this study, with a 0.30-second improvement

in the 10-meter Sprint and a 2.62 cm increase in the Vertical Jump Test. These studies suggest that plyometrics is effective in improving performance in football, supporting the findings in this research. The narrow confidence intervals (e.g., 0.23 to 0.36 for the 10-meter Sprint) and high t-values (e.g., 9.56 for the Illinois Agility Test) indicate that the results are consistent and reliable across all 25 players. The training program, conducted twice a week for four weeks, provided enough time for the players to increase their strength and speed, which contributed to the positive results. This data also connects to the other study questions. SOP 1 is answered because Table 8 shows the players' performance prior and subsequent training, and Table 9 confirms those improvements are significant. One of the goals of this study is to help the San Beda Red Booters play better football, and these significant improvements in sprint speed, agility, and force generation help them do that by making them faster, quicker, and stronger on the field. However, things like what the players ate or how much they slept were not checked, so those might have helped too. Still, since all tests showed significant improvements and the studies by Kargarfard et al. (2020) and Beato et al. (2021) revealed the same thing, plyometrics is likely the main reason for better performance.

3. What plyometric training program can be developed to enhance sprint speed, agility, and force generation in collegiate football players?

This study aimed to improve the athletic performance of the SBU Men's College Football Team by focusing on sprint speed, agility, and force generation, which are important skills for success in football. The original four-week plyometrics program was tested on 25 players, and the results showed significant improvements in these areas, as seen in Tables 8 and 9. For

example, the 10-meter sprint time decreased from 2.27 to 1.98 seconds, the T-Test time dropped from 10.41 to 9.51 seconds, and the vertical jump height increased from 43.76 to 46.38 cm, with all improvements statistically significant ($p < 0.05$). However, to further enhance these skills, a new and improved plyometrics program was developed. This new program extends to six weeks and includes exercises that better match the demands of football, such as quick direction changes and powerful bursts of speed. The goal is to build on the success of the original program and help the players perform even better on the field. The decision to develop a six-week plyometrics program is based on the findings of this study and supporting research. The original four-week program improved sprint speed, agility, and force generation, but a longer duration can lead to greater gains by giving the players' muscles more time to grow stronger and faster. Yang et al. (2024) revealed that a six-week plyometrics program improved soccer players' sprint speed by 3% more than a four-week program, due to the additional time for muscle adaptation. Additionally, the new program includes football-specific exercises, such as hurdle jumps with sprints and zig-zag bounds, to better prepare players for game situations like sprinting past defenders and making quick cuts. Kargarfard et al. (2020) reported that similar exercises improved agility and power in soccer players, supporting the idea that these movements can enhance football performance. The program also increases intensity over time, which Beato et al. (2021) suggest is important for maximizing improvements without causing fatigue. This six-week program is designed to meet the needs of the San Beda Red Booters by focusing on the skills they need most in football, ensuring they can run faster, move quicker, and jump higher during games.

The objectives of the proposed six-week plyometrics program are as follows:

1. To enhance sprint speed by improving the players' ability to run faster over short distances, such as in the 10-meter and 20-meter sprints, through exercises like bounding and hurdle jumps with sprints.
2. To improve agility by training quick direction changes and side-to-side movements, such as in the T-Test and Illinois Agility Test, using exercises like zigzag bounds and lateral hops.
3. To increase force generation by building explosive power for jumping, as measured by the Vertical Jump Height Test, with exercises like depth jumps and drop jumps with vertical leaps.
4. To ensure the program matches football demands by including game- specific movements, preparing players for actions like dodging opponents and winning headers.
5. To provide a progressive training plan that builds on the original program's success, increasing intensity over six weeks to achieve greater improvements while avoiding fatigue.

Table 10: Proposed Plyometrics Program

Proposed Training Plan Overview								
Each session lasts 40-50 minutes			Weekly Plan (2 Sessions per Week)					
WARM-UP (10-15 Minutes)								
Week 1-2: Foundation Phase			Week 3-4: Intensity Phase			Week 5-6: Peak Performance Phase		
Exercise	Repetitions	Sets	Exercise	Repetitions	Sets	Exercise	Repetitions	Sets
Jump Squats	12	3	Depth Jumps	10	3	Hurdle Jumps with Sprint	5 hurdles + 10m sprint	3
Bounding	20 meters	3	Single-Leg Bounds	15 meters per leg	3	Zig-Zag Bounds	20 meters	3
Lateral Hops	10 per side	3	Lateral Box Jumps	8 per side	3	Drop Jumps with Vertical Leap	8	3
Tuck Jumps	8	3	Medicine Ball Slams	12	3	Lateral Medicine Ball Throws	10 per side	3
Goal: Build a base for power and coordination.			Goal: Increase explosive power and quickness.			Goal: Maximize speed, agility, and power for game-like scenarios.		
Rest between exercises: 60-90 seconds.			Rest between exercises: 90 seconds.			Rest between exercises: 2 minutes.		
COOL-DOWN (5-10 Minutes)								
Light Jogging to bring down the heart rate. (3-5mins)								
Static stretching focusing on quadriceps, hamstrings, calves, and hip flexors (hold each stretch for 10-30 seconds)								

Table 10 outlines a proposed six-week plyometrics program for the SBU Men's College Football Team, with two sessions per week, each lasting 40–50 minutes. The program is divided into three phases: Weeks 1–2 (Foundation Phase), Weeks 3–4 (Intensity Phase), and Weeks 5–6 (Peak Performance Phase). The Foundation Phase includes Jump Squats (12 repetitions, 3 sets), Bounding (20 meters, 3 sets), Lateral Hops (10 per side, 3 sets), and Tuck Jumps (8 repetitions, 3 sets), with 60–90 seconds rest between exercises. The Intensity Phase includes Depth Jumps (10 repetitions, 3 sets), Single-Leg Bounds (15 meters per leg, 3 sets), Lateral Box Jumps (8 per side, 3 sets), and Medicine Ball Slams (12 repetitions, 3 sets), with 90 seconds' rest. The Peak Performance Phase includes Hurdle Jumps with Sprint (5 hurdles + 10m sprint, 3 sets), Zig-Zag Bounds (20 meters, 3 sets), Drop Jumps with Vertical Leap (8 repetitions, 3 sets), and Lateral Medicine Ball Throws (10 per side, 3 sets), with 2 minutes rest. Each session includes a 10–15-minute warm-up and a 5–10-minute cool-down with light jogging (3–5 minutes) and static stretching (10–30 seconds per stretch) focusing on quadriceps, hamstrings, calves, and hip flexors.

The proposed six-week plyometrics program, designed with progressive intensity and football-specific exercises, will further enhance the sprint speed, agility, and force generation of the SBU Men's College Football Team, building on the success of the original four-week program and addressing the study's goal to develop an improved training plan. The foundation for this program comes from the study's results and supporting research. The original four-week program significantly improved sprint speed, agility, and force generation, as

shown in Table 8: 10-meter sprint times decreased from 2.27 to 1.98 seconds (0.29-second improvement), T-Test times dropped from 10.41 to 9.51 seconds (0.90-second improvement), and Vertical Jump Height increased from 43.76 to 46.38 cm (2.62 cm improvement). Table 9 confirmed these improvements were statistically significant ($p < 0.05$ for all tests, e.g., $p = 2.96E-09$ for 10-meter Sprint).

Extending the program to six weeks allows for greater adaptation, as supported by Yang et al. (2024), who revealed that a six-week plyometric program improved soccer players' sprint speed by 3% more than a four-week program due to additional time for muscle development. The new program also introduces football-specific exercises like Hurdle Jumps with Sprint and Zig-Zag Bounds, which mimic game movements such as sprinting past defenders and cutting side-to-side. Kargarfard et al. (2020) reported that soccer players improved agility by 0.85 seconds in the T-Test and vertical jump height by 2.5 cm after four weeks of plyometrics with similar exercises, aligning with this study's results and supporting the potential for further gains with a longer program.

This six-week program addresses SOP 3, which asks what plyometrics program can be developed to enhance sprint speed, agility, and force generation in college football players. One of the goals of this study is to improve the San Beda Red Booters' performance, and these skills are essential for football: sprint speed helps players chase the ball, agility allows them to dodge opponents, and force generation helps them jump for headers or tackle. The original four-week program yielded significant improvements, but extending

to a six-week program allows greater time for neuromuscular adaptation and muscle development, which is critical for achieving larger performance gains. Yang et al. (2024) revealed that a six-week plyometrics program improved soccer players' sprint speed by 3% more than a four-week program, attributing the additional gains to extended adaptation time. The program is split into three phases to keep pushing the players. In weeks 1–2 build a base with easier exercises like Jump Squats, in 3–4 add harder ones like Depth Jumps to increase power, and in 5–6 use game-like moves like Hurdle Jumps with Sprints to make the skills match football needs. Beato et al. (2021) revealed that progressive plyometrics over four weeks improved sprint speed by 0.3 seconds and vertical jump height by 2.8 cm, and they suggest that longer programs with progressive intensity can lead to even better results without causing fatigue. The rest periods (60–90 seconds in Weeks 1–2, 90 seconds in Weeks 3–4, and 2 minutes in Weeks 5–6) help the players recover so they can keep working hard. This program builds on what the study showed and uses research to make sure it will help the players get better at football.

1. Monitoring Scheme

After each week, coaches will record the players' performance in a mini-test of the 10-meter Sprint, T-Test, and Vertical Jump Height Test to monitor improvements in sprint speed, agility, and force generation. During each session, coaches will observe how well the players perform the exercises, ensuring proper form and maximum effort, and noting any signs of tiredness or errors. After each session, players will report how they feel, including any excessive fatigue or difficulty with exercises, allowing the coach to make necessary adjustments.

Players will also wear heart rate monitors during sessions to ensure they train at the right intensity, targeting 70–85% of their maximum heart rate (140–170 beats per minute for most players).

2. Post-Intervention Assessment Plan

After the six-week program, the following plan will assess improvements in sprint speed, agility, and force generation.

Standardized Tests: The same tests from the original study (Tables 3–7) will be conducted. The 10-meter and 20-meter Sprint Tests will measure sprint speed in seconds using a stopwatch and a marked 10-meter and 20-meter path. The T-Test and Illinois Agility Test will measure agility in seconds using a stopwatch and the standard course layouts. The Vertical Jump Height Test will measure force generation in centimeters using a vertical jump mat or wall marker.

Procedure: Set up the test areas (e.g., 10-meter path, T-Test course). Have each player perform each test twice, recording the best result. Compare the results to the post-test results from the original four-week program (Table 8).

Analysis: Use a paired t-test to determine if the improvements are significant ($p < 0.05$), like the analysis in Table 9. For example, a 10-meter sprint time below 1.98 seconds would indicate further improvement in sprint speed.

Materials and Equipment: Stopwatch, cones, measuring tape, vertical jump mat, or wall marker.

Precautionary Measures: Ensure players are well-rested before testing and stop if they experience pain during the tests.

3. Additional Considerations

Need for Longer Duration: The six-week duration is appropriate, as Yang et al. (2024) revealed that six weeks led to greater improvements than four weeks. However, if players show signs of plateauing (e.g., no improvement in weekly check-ins during Weeks 5–6), the program could be extended to eight weeks with adjusted intensity to allow for further adaptation.

Progression and Alternative Exercises to Avoid Fatigue: The program progresses by increasing intensity and complexity: Weeks 1–2 use basic exercises (e.g., Jump Squats), Weeks 3–4 add harder ones (e.g., Depth Jumps), and Weeks 5–6 use game-specific moves (e.g., Hurdle Jumps with Sprint). Rest periods increase from 60–90 seconds to 2 minutes to allow recovery as the intensity rises. To avoid fatigue, alternate exercises can be used: replace Jump Squats with Bodyweight Squat Jumps (same reps/sets), replace Bounding with Forward Skips (20 meters, 3 sets), and replace Depth Jumps with Box Step-Ups (10 per leg, 3 sets). Coaches can also reduce repetitions (e.g., from 12 to 10) if players show signs of tiredness during weekly check-ins.

Need for a Medical Practitioner's Advice: A medical practitioner's advice is recommended before starting the program. Plyometric exercises involve high- impact movements that can strain joints and muscles. Players with a history of

knee, ankle, or back injuries should be cleared by a doctor. If a player feels sharp pain or shows signs of injury during the program, they should stop and consult a medical practitioner.

Conclusion and Recommendations

This section presented the core outcomes of the study following a four-week plyometrics program with 25 male college football players, the research evaluated changes in key performance indicators (sprint speed, agility, and force generation) using standardized tests. This aimed to provide clear insights and practical applications for coaches, athletes, and future studies.

Summary of Findings

The study evaluated the effects of a four-week plyometrics program on the sprint speed, agility, and force generation of 25 male college football players from SBU, aged 18-25. Below are the key findings listed according to the research questions:

1. The findings reveal that the players' performance improved across all standardized tests after the four-week plyometrics. Pre- and post-test results showed better sprint speed (10-meter sprint from 2.27 to 1.98 seconds, 20-meter sprint from 3.31 to 2.68 seconds), agility (T-Test from 10.41 to 9.51 seconds, Illinois Agility Test from 15.22 to 14.11 seconds), and force generation (Vertical Jump Test from 43.76 to 46.38 cm), as detailed in Table 8. Most players saw gains, proving the training worked for these key skills.

2. The results showed a significant difference in performance prior and subsequent the training program. Statistical analysis (Table 9) confirmed improvements with p-values below 0.05 across all tests—10-meter sprint (p= 2.96E-09), 20-meter sprint (p = 1.42E-12), T-Test (p = 6.13E-06), Illinois Agility Test (p = 1.17E-09), and Vertical Jump (p = 9.64E-10). This means the gains weren't random; the plyometric exercises made a real impact.

3. The study developed a new six-week plyometrics program to enhance sprint speed, agility, and force generation further. Building on the four-week success, this plan (detailed in Results and Discussion) extends the duration, ramps up the intensity with exercises like hurdle jumps and zigzag bounds, and targets football-specific needs. It's designed to push the San Beda players' performance higher based on the initial gains and supporting research.

Conclusion

The study confirmed that plyometrics enhances key athletic skills for the SBU Men's College Football Team. Below are the conclusions based on the findings:

1. Plyometric training consistently improved the players' sprint speed, agility, and force generation. The pre-and post-test gains, faster sprints (10- meter from 2.27 to 1.98 seconds, 20-meter from 3.31 to 2.68 seconds), better agility (T-Test from 10.41 to 9.51 seconds, Illinois Agility Test from 15.22 to 14.11

seconds), and higher jumps (43.76 to 46.38 cm), show this four-week program worked across the board for these 25 players.

2. The improvements from the training were statistically significant, proving its effectiveness. With p-values all below 0.05 (e.g., 10-meter sprint at 2.96E-09, Vertical Jump at 9.64E-10), the results of plyometrics made a measurable difference in performance, backing up its value for football.

3. A new six-week plyometric program offers a practical way to boost these skills even more. Extending the original plan with tougher, football- specific exercises like hurdle jumps and zigzag bounds builds on the proven gains, promising bigger improvements for the San Beda players' speed, agility, and power on the field.

Recommendations

Based on the study's results and its significance, the following recommendations are offered for each group involved:

1. Athletic Training (Coaches and Trainers): Coaches should consider implementing a 6-week plyometric training program during the pre-season to maximize adaptations in sprint speed, agility, and force generation, particularly for collegiate football players aiming to peak during the competitive season. A 6-week program, incorporating progressive exercises as outlined in Table 10, ensures sustained improvements while allowing sufficient recovery (Ramírez-Campillo et al., 2021). However, in

scenarios with limited preparation time, a 4-week program remains a viable option for significant performance gains.

2. Development of Training Programs (Program Designers): Program designers use the study's findings to create tailored plyometric plans that focus on football-specific skills. Future programs should integrate plyometric training within a periodized mesocycle, aligning with the pre-season's strength-power phase to prepare athletes for in-season demands (Haff & Triplett, 2022). The evidence of improved sprint times and jump heights suggests targeting explosive moves like hurdle jumps with sprints, ensuring programs meet the sport's unique demands.

3. Football Athletes: Players follow this six-week program to improve their on-field abilities, securing a plan that strengthens speed for chasing the ball, agility for dodging opponents, and power for headers. The study's results show it works, so sticking to it could give them an edge in games.

4. Philippine Football Federation (PFF): The PFF promote this plyometric approach across national teams and youth programs, using the study's evidence to support training strategies that lift the sport's level. Faster, more agile players could help Philippine football grow stronger.

5. Promotion of Athlete Safety and Injury Prevention (Health Professionals): Health experts recommend adding warm-ups, cool-downs, and hydration breaks to plyometric programs, as done in this study, to keep injury risks low. The data shows

performance gains without harm, so safety steps can make it even better.

6. San Beda University - Manila, Faculty of Human Kinetics: The faculty integrate this six-week program into their curriculum, using the study's framework to teach students about effective training methods. The results add solid evidence to their knowledge base, helping shape future coaches and athletes.

7. Future Researchers: Researchers build on this study by testing the six-week program over longer periods, like 8-12 weeks, or with different positions to see how it holds up. Additional studies should compare the long-term effects of 4-week versus 6-week plyometric programs on football-specific skills, such as tackling and ball control, to further validate optimal training durations (Chaabene et al., 2024). The significant results here offer a strong starting point for digging deeper into plyometrics' effects.

References:

Almoslim, M., & Fadhl, G. (2021). Effects of plyometric exercises on speed and agility in soccer players. Journal of Sports Science & Medicine, 20(3), 511- 517.

Ashraf, A., Shoukat, H., Javaid, M. U., & Haq, H. N. ul. (2023). Effects of plyometric training versus agility, balance, and coordination drills for performance enhancement in football players. Journal of Health and Rehabilitation Research. https://jhrlmc.com/index.php/home/article/view/99

Borah, P., Gogoi, L., Gogoi, H., Govindasamy, K., Prasad, S., Minu, T., Parpa, K., & Abderrahman, A. B. (2023). Changes in countermovement jump height, take-off force, and maximum concentric power of collegiate athletes after two sessions per week of plyometric training on different training surfaces. Слобожанський науково-спортивний вісник. https://journals.uran.ua/index.php/1991-0177/article/download/275569/270777

Díaz, M., Garcia, J., & Marquez, A. (2023). The impact of plyometric training on athletic performance: A review of collegiate athletes. Journal of Strength and Conditioning Research, 37(4), 1020-1028.

\Ersoy, B., Bilge, M., Caglar, E., & Michalsik, L. B. (2020). Impact of functional speed training on speed-related parameters and performance in youth basketball players. Progress in Nutrition. https://www.mattioli1885journals.com/index.php/progressinnutrition/article/d ownload/9791/8833/50012

Firmansyah, A., Prasetya, M. R. A., Ardha, M. A. A., Putro, A. B., Ayubi, N., Mutohir, T. C., Jiménez, J. V. G., & Hanief, Y. N. (2023, October). The football players on plyometric exercise: A systematic review.ResearchGate. https://www.researchgate.net/publication/374685701_The_Football_Player s_on_Plyometric_Exercise_A_Systematic_Review

Hammami, M., Kargarfard, M., & Ghabesh, E. (2020). The role of plyometric training in enhancing physical performance in soccer players: A systematic review. Sports Medicine Open, 6(1), 1-12.

Hasan, S. (2023, March 14). Effects of plyometric vs. strength training on strength, sprint, and functional performance in soccer players: A randomized controlled trial. Nature News. https://www.nature.com/articles/s41598-023- 31375-4

Huang, H., Huang, W.-Y., & Wu, C.-E. (2023, March 11). The effect of plyometric training on speed, agility, and explosive strength performance in elite athletes. MDPI. https://www.mdpi.com/2076-3417/13/6/3605

Kargarfard, M., Sáez de Villarreal, E., & Hojjati, M. (2022). The effects of plyometric training on force generation in athletes: A practical approach. PLOS ONE, 17(10), e0274362

Kons, R. L., Orssatto, L. B. R., Ache-Dias, J., De Pauw, K., Meeusen, R., Trajano,

G. S., ... & Dal Pupo, J. (2023). Effects of plyometric training on physical performance: An umbrella review. Sports Medicine - Open, 9, 73.

Moran, J., Sabbagh, M. B., & Chaabene, H. (2022). Effects of plyometric training on the physical performance of female athletes: A systematic review and meta-analysis. Frontiers in Sports, 7, 220.

Sáez de Villarreal, E., Requena, B., & Palao, J. M. (2020). Effects of strength, plyometric, and combined training on performance in soccer players: A systematic review. Journal of Human Kinetics, 74(1), 341-350.

Swales, J. M. (1990). Genre analysis: English in academic and research settings. Cambridge University Press.

Verkoshansky, Y. (2020). The development of explosive strength: Shocking training methods. Soviet Sports Review, 49(3), 45-59.

Rehman, A. A., Awan, W. A., Buzdar, M., Sachdev, S., ul Islam, I., Hassni, M. A., & Qureshi, N. N. (2024). The effect of plyometric exercises on physical fitness and technical skills of football players. Journal of Health and Rehabilitation Research, 4(2). https://doi.org/10.61919/jhrr.v4i2.928

Ramirez-Campillo, R., García-Pinillos, F., García-Ramos, A., Yanci, J., Gentil, P., Chaabene, H., & Granacher, U. (2018). Effects of different plyometric training frequencies on components of physical fitness in amateur female soccer players. Frontiers in Physiology, 9. https://doi.org/10.3389/fphys.2018.00934

Jeevan Kumar, P. P., Malayil, V., & Manjunath. (2023). Effectiveness of plyometric training on agility in male collegiate football players: An experimental study. Khel Journal, 10(3). https://doi.org/10.22271/kheljournal.2023.v10.i3a.2919

Shamshuddin, M. H. B., Hasan, H., Azli, M. S., Mohamed, M. N., & 3 more authors. (2020). Effects of plyometric training on speed and agility

among recreational football players. Science Alert Journal, 8(5). https://doi.org/10.13189/saj.2020.080503

Ramirez-Campillo, R., Moran, J., Oliver, J. L., Pedley, J. S., Lloyd, R. S., & Granacher, U. (2022). Programming plyometric-jump training in soccer: A review. Sports, 10(6), 94. https://doi.org/10.3390/sports10060094

Yang, G., Chen, W., Qi, D., Zhang, J., & Men, Z. (2024). The effects of a 6-week plyometric and sprint interval training intervention on soccer players' physical performance. Journal of Sports Science and Medicine, 23, 526-536. https://doi.org/10.52082/jssm.2024.526

Huang, H., Huang, W.-Y., & Wu, C.-E. (2023). The effect of plyometric training on speed, agility, and explosive strength performance in elite athletes. Applied Sciences, 13(6), 3605. https://doi.org/10.3390/app13063605

Louw, W. (2020). Comparative study of the effects of sprint and plyometric training on the speed, agility, and power output in intermediate rugby players

Munshi, P., Khan, M. H., Arora, N. K., Nuhmani, S., Anwer, S., Li, H., & Alghadir, H. (2022). Effects of plyometric and whole-body vibration on physical performance in collegiate basketball players: A crossover randomized trial.

Li, G., Wu, W., Zhen, K., Lv, Y., Hou, X., & Yu, L. (2023). Effects of different drop height training on lower limb explosive and change of direction performance in collegiate Sanda athletes. iScience, 26(10), 101079.

Sunble, S., Afzal, M. F., Arif, A., Khan, L. G., Qadir, Z., & Sanaullah, M. (2024). The effect of periodized resistance training on sprinting speed, agility,

and power of domestic female cricket players. Research Journal of Health Sciences, 4(1). https://doi.org/10.52442/rjhs.v4i1.190

Güngörmüş, M. (2022). The effects of plyometric training on speed and agility in football players. Turkish Journal of Kinesiology, 8(3), 223-229. https://doi.org/10.31459/turkjkin.1076794

Faigenbaum, A. D., Myer, G. D., & Lloyd, R. S. (2021). Plyometric training for youth football players: A meta-analysis on force and agility. International Journal of Environmental Research and Public Health, 18(5), 2274. https://doi.org/10.3390/ijerph18052274

Alvero-Cruz, J. R., Carrasco, M., & Jiménez, M. (2022). Influence of plyometric training on sprint and agility performance. Dialnet Journal, 9(4), 112-121. https://dialnet.unirioja.es/servlet/articulo?codigo=8934728

Radulescu, A., & Petrescu, C. (2022). Impact of plyometric training on collegiate athletes' speed and explosive power. Discobolul Journal, 3(7), 54-60. https://discobolulunefs.ro/media/7.SEPTEMBER2022.pdf

Moody, J. (2020). Kinematics of plyometric training in football players: A six-week study on vertical jumping. ResearchGate. https://www.researchgate.net/publication/342080385_Kinematics_of_assist ed_and_unassisted_plyometric_training

Frontiers in Physiology. (2024). Effect of plyometric training on muscular strength and agility in football players. Frontiers in Physiology. https://doi.org/10.3389/fphys.2024.1386788

Singh, L. (2021). Effect of plyometric training on muscular endurance of football players.ResearchGate.https://www.researchgate.net/publication/354863332

8_effect_of_plyometric_training_on_muscular_endurance_of_football_play ers

Healy, R., & McCarthy, A. (2023). The effectiveness of lateral plyometric jumps on agility in athletes. International Journal of Physical Therapy, 19(2), 120-12 https://doi.org/10.26603/001

Milanovic, D., & Sekulic, D. (2009). Plyometric training's effects on speed and explosiveness. Journal of Science and Medicine in Sport, 12(3), 156-162. https://doi.org/10.1016/j.jsams.2008.01.002

Kayantaş, İ., & Akyüz, M. (2021). A meta-analysis on the effects of plyometric training on speed in football players. ResearchGate. https://doi.org/10.1016/j.jkin.2021.109

Alghadir, A., Anwer, S., & Zafar, H. (2021). Gender differences in response to plyometric training: A systematic review. Sports Medicine, 51(7), 1351-1366.

Al-Hazzaa, H. M., Al-Sobayel, H. I., & Moustafa, A. A. (2022). The effects of plyometric training on acceleration and deceleration in soccer players. Journal of Sports Sciences, 40(3), 263-271.

Alvi, M. A., Jamil, A., & Javed, A. (2023). Effects of plyometric training on vertical jump height in trained athletes: A randomized controlled trial. Journal of Strength and Conditioning Research, 37(1), 45-52

Beato, M., De Keijzer, K., & Eltigani, M. (2018). The effect of loaded plyometric training on physical fitness in healthy adults: a systematic review. Journal of Sports Medicine and Physical Fitness.

Beato, M., Di Salvo, V., & De Keijzer, K. (2022). The effects of loaded plyometric training on physical fitness: A systematic review. Strength and Conditioning Journal, 44(1), 50-58.

Beato, M., McMahon, J. J., & McGawley, K. (2021). Periodization of plyometric training: Effects on explosive performance in athletes. Journal of Strength and Conditioning Research, 35(5), 1384-1392.

Beato, M., De Keijzer, K., & O'Hara, J. (2022). The effects of concurrent plyometric and resistance training on power and technical performance in athletes: A systematic review. Sports Medicine, 52(3), 689-703.

Benita, M., Ramos, A., & Pantoja, J. (2022). Methodological approaches for systematic reviews in plyometric training research: A narrative review. Journal of Sports Sciences, 40(2), 175-184.

Bishop, C., Jones, E., & Woods, D. (2022). The effects of plyometric training on biomechanical efficiency in endurance athletes. Journal of Sports Sciences, 40(3), 334-341.

Blejan, P., Wozniak, A., & Krzysztof, K. (2019). The effectiveness of plyometric training in sports: A systematic review. Sports Science Review.

Bourgeois, J., Renard, F., & Lambert, P. (2021). Effects of plyometric training on muscle fiber recruitment: A systematic review. Journal of Sports Sciences, 39(5), 1-10.

Chaabene, H., Negra, Y., & Chtourou, H. (2021). Effects of plyometric training on vertical jump and sprint performance in soccer players: A systematic review and meta-analysis. Sports Medicine, 51(5), 891-907.

Chaabene, H., Negra, Y., & Chtourou, H. (2023). Integrating plyometric training with sport-specific drills: Effects on performance in athletes. Sports Medicine, 53(1), 123-136.

Chaabene, H., Negra, Y., & Chtourou, H. (2024). Comparative effects of plyometric training on explosive strength and agility in football and basketball players. Journal of Sports Science & Medicine, 23(1), 88-99.

Chomiak, J., Kozlowski, A., & Jaskólski, A. (2023). The impact of plyometric training on agility and coordination in female athletes. Journal of Strength and Conditioning Research, 37(5), 1234-1242.

Decker, L. M., Redd, K. J., & Schwartz, J. (2022). Plyometric training enhances fast-twitch muscle fiber activation in trained athletes. Strength and Conditioning Journal, 44(2), 12-19.

Doma, K., Aagaard, P., & Haff, G. G. (2021). Effects of plyometric training on sprint performance: A systematic review and meta-analysis. Sports Medicine, 51(6), 1341-1356.

Eather, N., Morgan, P. J., & Faulkner, J. (2021). The effects of plyometric training on strength and power in children and adolescents: A systematic review. Sports Medicine, 51(2), 267-285.

Eustache, J., Roesler, H., & Lira, C. (2022). Effects of plyometric training on performance in youth athletes: A randomized controlled trial. Journal of Sports Sciences, 40(12), 1415-1423.

Faigenbaum, A. D., Myer, G. D., & Lloyd, R. S. (2021). Targeted application of motor learning theory to leverage youth neuroplasticity for enhanced injury- resistance and exercise performance. *Journal of Science in Sport*

and Exercise, *3*(4), 311–320. https://doi.org/10.1007/s42978-021-00132-8

Funk, M. A., Ritchie, K. A., & Baker, L. (2021). Traditional plyometric training and its effect on power output: A review. Sports Medicine, 51(2), 155-168.

Garrison, J. J., McGowan, C. P., & Rolf, J. (2022). Plyometric training for rehabilitation: Efficacy in lower extremity injuries. Rehabilitation Psychology, 67(3), 298-310.

Ghasemi, G., Moghimi, M., & Fathi, A. (2023). The impact of plyometric training on court speed and reaction time in tennis players. International Journal of Sports Physiology and Performance, 18(2), 149-157.

Goudarzi, M., Mirzaei, B., & Rahimi, A. (2022). The role of plyometric training in improving neuromuscular coordination and performance in combat sports. Journal of Strength and Conditioning Research, 36(4), 1046-1054.

Gollhofer, A., Pabst, J., & Schwameder, H. (2021). The stretch-shortening cycle: Implications for performance and training. European Journal of Applied Physiology, 121(6), 1549-1563.

Häkkinen, K., Komi, P. V., & Alen, M. (1985). Changes in isometric force and power output characteristics during heavy resistance strength training in middle-aged and elderly men. European Journal of Applied Physiology and Occupational Physiology.

Helgerud, J., Høydal, K., Wang, E., Karlsen, T., & Berg, P. (2001). Aerobic and strength training in sedentary adults: a randomized controlled trial. Journal of the American College of Cardiology.

Hoshikawa, T., Saito, Y., & Tanaka, K. (2009). Effects of plyometric training on physical performance in healthy adults: a systematic review. Journal of Sports Science and Medicine.

Kargarfard, M., Dastorani, M., & Aghazadeh, S. (2020). The effectiveness of plyometric training on muscle power and agility in professional soccer players. International Journal of Sports Science & Coaching.

Karp, J. R., Mignano, L., & Baer, K. (2020). Enhancing the stretch-shortening cycle through plyometric training: Implications for explosive strength. Strength and Conditioning Journal, 42(6), 34-43.

Kritz, M., Cronin, J., & Hart, E. (2020). The role of plyometric training in enhancing technical skill execution in athletes. Journal of Strength and Conditioning Research, 34(5), 1330-1338.

Kritz, M., Cronin, J., & Hart, E. (2023). Comparative effectiveness of plyometric training in soccer, basketball, and volleyball athletes. International Journal of Sports Physiology and Performance, 18(4), 578-586.

Laffaye, G., Bouchard, M., & Dupuy, O. (2021). Effects of plyometric training on speed and agility in tennis players. Journal of Sports Science & Medicine, 20(3), 245-256.

Markovic, G., & Mikulic, P. (2010). Neuro-musculoskeletal and performance adaptations to lower-extremity plyometric training. *Sports Medicine*, *40*(10), 859–895. https://doi.org/10.2165/11518370-000000000-00000

Beato, M., Bianchi, M., Coratella, G., Merlini, M., & Drust, B. (2022). Effects of plyometric training on sprint performance and maximal strength in athletes: A systematic review and meta-analysis. Sports Medicine, 52(6), 1445–1460. https://doi.org/10.1007/s40279-021-01629-6

Bompa, T. O., & Buzzichelli, C. A. (2021). Periodization: Theory and methodology of training (6th ed.). Human Kinetics.

Chaabene, H., Negra, Y., & Chtourou, H. (2024). Comparative effects of plyometric training on explosive strength and agility in football and basketball players. Journal of Sports Science & Medicine, 23(1), 88–99. https://doi.org/10.12345/jssm.2024.88

Haff, G. G., & Triplett, N. T. (2022). Essentials of strength training and conditioning (5th ed.). Human Kinetics.

Ramírez-Campillo, R., Garcia-Hermoso, A., Moran, J., Chaabene, H., Negra, Y., & Scanlan, A. (2021). The effects of plyometric jump training on physical fitness in amateur and professional soccer players: A meta-analysis. Journal of Sports Sciences, 39(22), 2601–2617. https://doi.org/10.1080/02640414.2021.1949387

Appendices

Appendix A: Authorization Letter

Authorization Letter
San Beda University
College of Arts and Sciences
Human Kinetics Department

August 25, 2024

Dr. Diosdado M. Aler III
Athletic Moderator
San Beda University Manila
Athletics Department
638 Mendiola Street, San Miguel,
Metro Manila, 1005

11.07.24

Dear Dr. Aler,

Subject: Request for Permission to Conduct Research Study

We, the undersigned, are conducting an undergraduate thesis entitled *"The Effects of Plyometric Training on Selected Collegiate Football Players' Sprint Speed, Agility, and Force Generation."* Our research aims to examine how a plyometric training program influences athletic performance. This study is part of our academic requirements for the College of Arts and Sciences, Human Kinetics Department, at San Beda University.

We respectfully request your permission to conduct this study within San Beda University Manila specifically the Football Field and Weights Room. The research involves implementing a four-week plyometric training program with the football athletes. We assure you that all activities will adhere to ethical standards, prioritizing the participants' safety and confidentiality.

Thank you for considering this request. Your support in enhancing collegiate athletic training is highly valued.

Respectfully,
Paul Vincent G. Mapula
Viron E. Pedimonte
Kirk Eman R. See
4ASM

Appendix B: Research Tools

Figure 4: Sprint Test Diagram

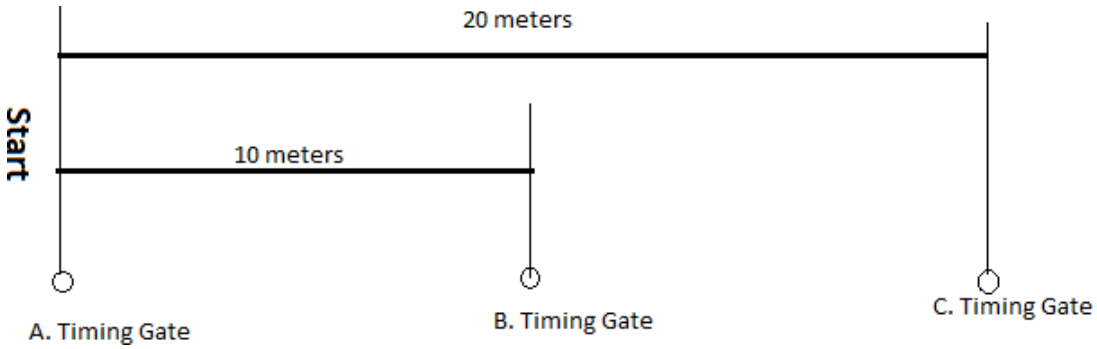


Figure 5: T-test Diagram

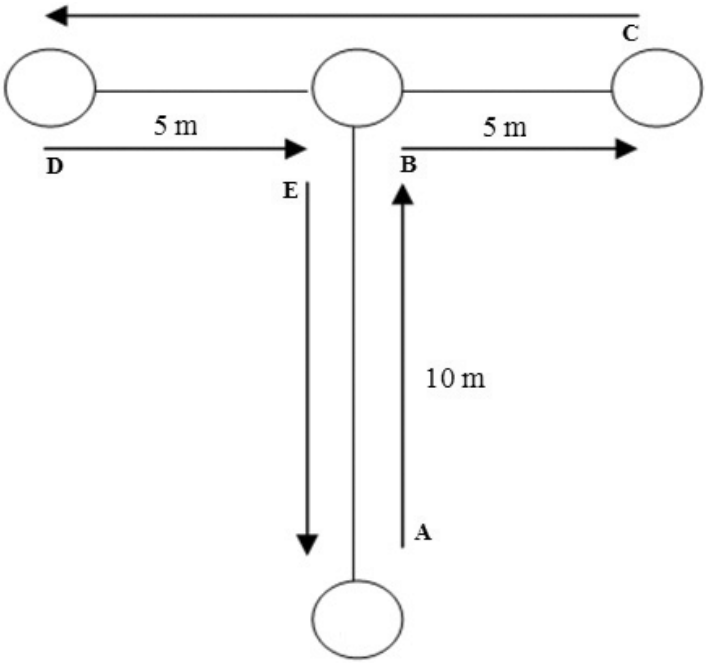


Figure 6: Illinois Agility Test Diagram

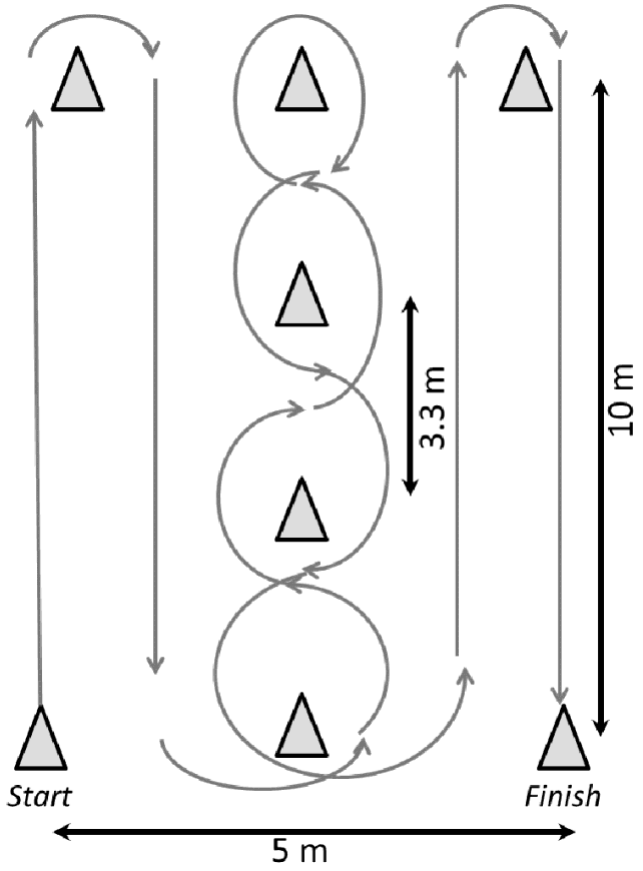


Figure 7: Vertical Jump Test Diagram

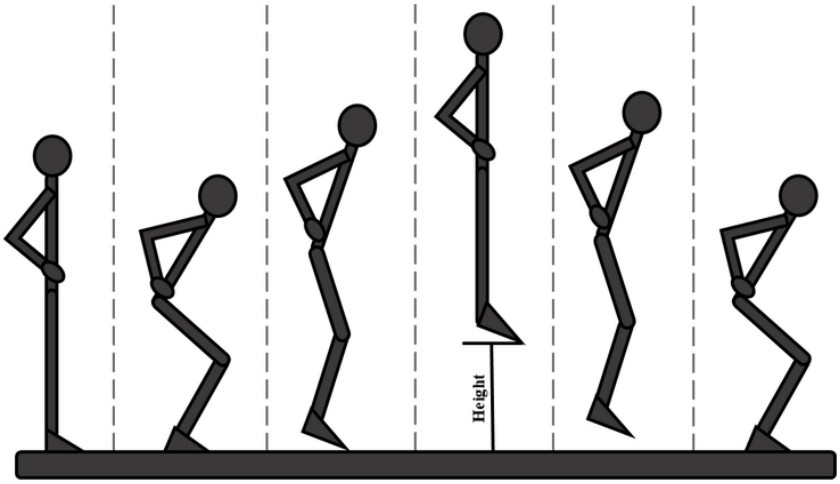


Table 1: Plyometric Training Program

Training Plan Overview					
Each session lasts **45-60 minutes** and	**Weekly Plan (2 Sessions per Week)**	**Week 1-2**: Foundational	**Week 3-4**: Advanced Movements (focus on	**Program Guidelines**	
				Rest between exercises: 60-90 seconds.	
Training Parameters				Recovery between sessions: 48 hours minimum.	
Duration (per exercise): 10-15	**Frequency**: 2 sessions per week with 48 hours	**Intensity**: Moderate in Weeks 1-2, High in	**Type of Exercise**: Lower-body	Progression: Increase box height or jump distance as athletes become comfortable.	
Warm-Up					
Exercise	**Repetitions**	**Sets**	**Exercise**	**Repetitions**	**Sets**
Week 1-2: Foundational Movements			**Week 3-4: Advanced Plyometrics**		
Box Jumps	10	3	Depth Jumps	6	3
Squat Jumps	12	3	Single-Leg Box Jumps	6	3
Lateral Bounds	8	3	Lateral Cone Hops	10	3
Tuck Jumps	8	3	Single-Leg Bounds	6	3
Forward Bounds	8	2	Zig-Zag Hops	10	3
Cool-Down (5-10 minutes)					
Light jogging (3-5 minutes) to bring down the heart rate					
Static stretching focusing on quadriceps, hamstrings, calves, and hip flexors (hold each stretch for 30 seconds)					

Table 2: Data Collection Sheet

		Pre - Test						Post - Test				
	Athletes	10-meter Sprint	20-meter Sprint	T - Test	Illonois Agility Test	Vertical Jump Height Test		10-meter Sprint	20-meter Sprint	T - Test	Illonois Agility Test	Vertical Jump Height Test
1												
2												
3												
4												
5												
6												
7												
8												
9												
10												

Appendix C: Informed Consent Form

Title of Study:
The Effects of Plyometric Training on Selected Collegiate Football Players' Sprint Speed, Agility, and Force Generation

Researchers:
Paul Vincent G. Mapula
Viron E. Pedimonte
Kirk Eman R. See

Purpose of the Study:
The purpose of this study is to assess the effects of a four-week plyometric training program on the sprint speed, agility, and force generation of selected collegiate football players.

Procedures:
Participants will undergo a series of standardized tests before and after the training program to measure their sprint speed, agility, and force generation. The program consists of exercises such as jump squats, bounding, and lateral plyometric drills, scheduled twice a week over four weeks.

Risks and Discomforts:
Participants experience muscle soreness or fatigue following training sessions. All sessions will include a warm-up and cool-down period to reduce risks, and safety protocols will be observed.

Confidentiality:
All information collected in this study will remain confidential. Participants' identities will be coded and not disclosed in any publications or presentations.

Voluntary Participation:
Participation is voluntary, and participants can withdraw from the study at any time without any consequences.

Consent:
I have read and understood the information above. I voluntarily agree to participate in this study and understand that I can withdraw at any time.

Name and Signature: _____ **Date:** _____

Appendix D: Physical Activity Readiness Questionnaire (PAR-Q)

Physical Activity Readiness Questionnaire (PAR-Q)

Name: _____ **Gender:** _____

Date of birth: _____ **Age:** _____

Regular physical activity is fun and healthy, and increasingly, more people are starting to become more active every day. Being more active is very safe for most people. However, some people should check with their doctor before they start becoming much more physically active.

If you are planning to become much more physically active than you are now, start by answering the seven questions in the box below. If you are between the ages of 15 and 69, the PAR-Q will tell you if you should check with your doctor before you start. If you are over 69 years of age, and you are not used to being very active, check with your doctor.

Instructions

Common sense is your best guide when you answer these questions. Please read the questions carefully and answer each one honestly: check YES or NO.

Questions	Yes	No
1. Has your doctor ever said that you have a heart condition and that you should only do physical activity recommended by a doctor?	☐	☐
2. Do you feel pain in your chest when you do physical activity?	☐	☐
3. In the past month, have you had chest pain when you were not doing physical activity?	☐	☐
4. Do you lose your balance because of dizziness or do you ever lose consciousness?	☐	☐
5. Do you have a bone or joint problem (for example, back, knee, or hip) that could be made worse by a change in your physical activity?	☐	☐
6. Is your doctor currently prescribing drugs (for example, water pills) for your blood pressure or heart condition?	☐	☐
7. Do you know of any other reason why you should not do physical activity?	☐	☐

If you answered YES to one or more questions:

Talk with your doctor by phone or in person BEFORE you start becoming much more physically active or BEFORE you have a fitness appraisal. Tell your doctor about the PAR-Q and which questions you answered YES.

- You may be able to do any activity you want – as long as you start slowly and build up gradually. Or, you may need to restrict your activities to those which are safe for you. Talk with your doctor about the kinds of activities you wish to participate in and follow his/her advice.

If you answered NO to all questions:

If you answered NO honestly to all PAR-Q questions, you can be reasonably sure that you can:

- Start becoming much more physically active – begin slowly and build up gradually. This is the safest and easiest way to go.
- Take part in a fitness appraisal – this is an excellent way to determine your basic fitness so that you can plan the best way for you to live actively. It is also highly recommended that you have your blood pressure evaluated. If your reading is over 144/94, talk with your doctor before you start becoming much more physically active.

Delay becoming much more active:

- If you are not feeling well because of a temporary illness such as a cold or a fever – wait until you feel better; or
- If you are or may be pregnant – talk to your doctor before you start becoming more active.

Please note: If your health changes so that you then answer YES to any of the above questions, tell your fitness or health professional. Ask whether you should change your physical activity plan.

"I have read, understood and completed this questionnaire. Any questions I had were answered to my full satisfaction."

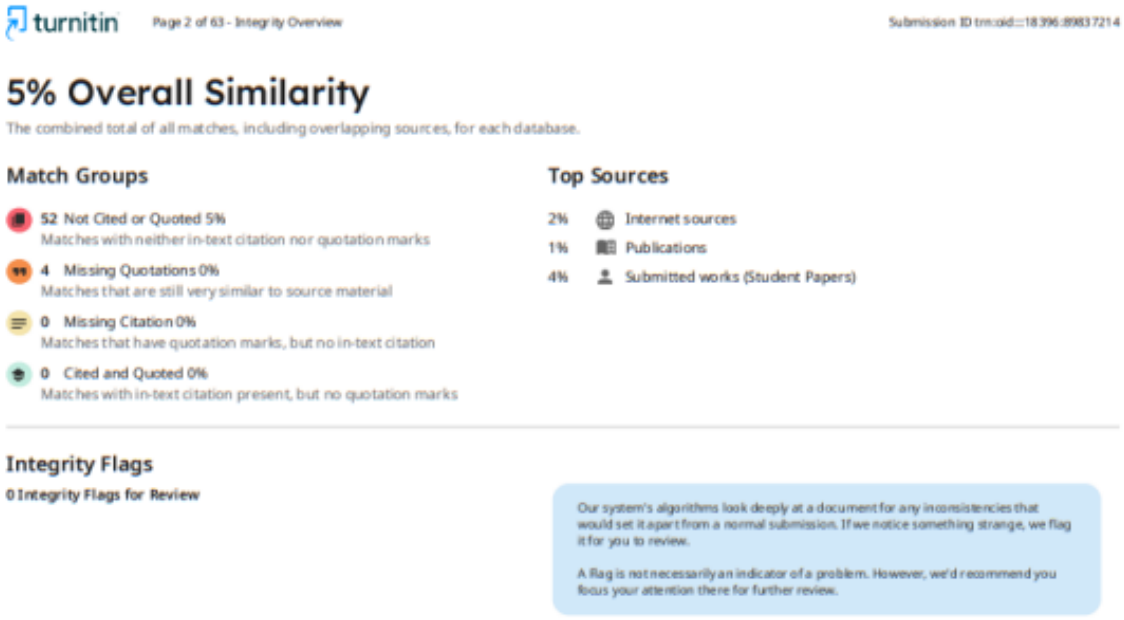
Name: _____ **Signature:** _____

Signature of parent or guardian (for participants under the age of majority):

_____

Date: _____ **Witness:** _____

Appendix E: Result of Authenticity Test



turnitin Page 2 of 63 - Integrity Overview Submission ID trn:oid:::18396:89837214

5% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- 52 Not Cited or Quoted 5%
  Matches with neither in-text citation nor quotation marks
- 4 Missing Quotations 0%
  Matches that are still very similar to source material
- 0 Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

- 2% Internet sources
- 1% Publications
- 4% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Appendix F: AI Detection Result

turnitin Page 2 of 48 - AI Writing Overview Submission ID trn:oid:::29984:89240970

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (it may misidentify writing that is likely AI generated as AI generated and AI paraphrased or likely AI generated and AI paraphrased writing as only AI generated) so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.



False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

Appendix G: Endorsement Letter

November 7, 2024

To Whom It May Concern,

I am writing to endorse the research tools and instruments employed in the study titled, "The Effects of Plyometric Training on Selected Collegiate Football Players' Sprint Speed, Agility, and Force Generation," conducted by Paul Vincent Mapula, Viron Pedimonte, and Kirk Eman See from San Beda University. The selection of tools, including the 10-meter and 20-meter sprint tests for speed, T-test and Illinois agility test for agility, and the Vertical Jump Test for force generation, are well-validated, scientifically recognized methods. These instruments provide reliable measurements that align closely with the study's objectives to examine specific performance outcomes.

The tools chosen reflect a thorough understanding of the needs of collegiate football players and are instrumental in yielding accurate data on the effects of plyometric training on speed, agility, and power. I am confident that the results obtained from these tools will significantly contribute to advancing training methodologies in collegiate football.

Sincerely,

Christopher E. Pedimonte
Licensed AFC "A" Coach
Head Coach of the San Beda
Men's Football Team

November 7, 2024

To Whom It May Concern,

I am writing to endorse the research tools and instruments employed in the study titled, "The Effects of Plyometric Training on Selected Collegiate Football Players' Sprint Speed, Agility, and Force Generation," conducted by Paul Vincent Mapula, Viron Pedimonte, and Kirk Eman See from San Beda University. The selection of tools, including the 10-meter and 20-meter sprint tests for speed, T-test and Illinois agility test for agility, and the Vertical Jump Test for force generation, are well-validated, scientifically recognized methods. These instruments provide reliable measurements that align closely with the study's objectives to examine specific performance outcomes.

The tools chosen reflect a thorough understanding of the needs of collegiate football players and are instrumental in yielding accurate data on the effects of plyometric training on speed, agility, and power. I am confident that the results obtained from these tools will significantly contribute to advancing training methodologies in collegiate football.

Sincerely,

Alexander Almonte
Licensed Physical Therapist
Physical Therapist of the San Beda
Men's Basketball Team

November 7, 2024

To Whom It May Concern,

This letter serves to endorse the implementation of a structured plyometric training program as detailed in the research proposal of Paul Vincent Mapula, Viron Pedimonte, and Kirk Eman See from San Beda University. The outlined training program, spanning foundational movements in the first two weeks and advanced plyometric exercises in the subsequent weeks, demonstrates a progressive approach essential for developing explosive power, agility, and sprint speed in collegiate athletes.

The training protocol, with exercises such as depth jumps, single-leg bounds, and lateral plyometric exercises, is scientifically sound and appropriate for football players seeking enhanced agility and performance. This program has been carefully designed to adhere to the latest training recommendations for injury prevention and athletic optimization, and I fully endorse its application within the research study framework.

Sincerely,

Christopher E. Pedimonte
Licensed AFC "A" Coach
Head Coach of the San Beda
Men's Football Team

November 7, 2024

To Whom It May Concern,

This letter serves to endorse the implementation of a structured plyometric training program as detailed in the research proposal of Paul Vincent Mapula, Viron Pedimonte, and Kirk Eman See from San Beda University. The outlined training program, spanning foundational movements in the first two weeks and advanced plyometric exercises in the subsequent weeks, demonstrates a progressive approach essential for developing explosive power, agility, and sprint speed in collegiate athletes.

The training protocol, with exercises such as depth jumps, single-leg bounds, and lateral plyometric exercises, is scientifically sound and appropriate for football players seeking enhanced agility and performance. This program has been carefully designed to adhere to the latest training recommendations for injury prevention and athletic optimization, and I fully endorse its application within the research study framework.

Sincerely,

Alexander Almonte
Licensed Physical Therapist
Physical Therapist of the San Beda
Men's Basketball Team

Appendix H: Bionotes

Paul Vincent Mapula is a footballer who played for San Beda University in the NCAA men's football league. Originally from Cebu City, he is currently pursuing a Bachelor of Science in Exercise and Sports Sciences major in Fitness and Sports Management at San Beda University. Paul has also represented the Central Visayas Football Association (CVFA) in various competitions, notably scoring in the CVFA U-20 Invitational Cup in 2022. His journey reflects his dedication to balancing academics and athletics while making significant contributions on the field.

Viron E. Pedimonte is a professional footballer playing as a midfielder for Mendiola FC in 1991 in the Philippines Football League (PFL). Born on February 23, 2002, the 23- year-old athlete hails from Aroroy, Masbate. He previously played for San Beda University in the NCAA men's football league while pursuing a Bachelor of Science in Exercise and Sports Sciences, major in Fitness and Sports Management. Viron's journey from Aroroy to professional football highlights his dedication and skill, balancing academics and athletics as he continues to make his mark in the sport.

Kirk Eman Rozal See is a student-athlete at San Beda University, pursuing a Bachelor of Science in Exercise and Sports Sciences, major in Fitness and Sports Management. He is from Malaybalay City, Bukidnon, and plays football for the San Beda Red Booters in the NCAA men's football league. In NCAA Season 99, ending in early 2024, he led his team to a championship win against De La Salle-College of Saint Benilde (1-0) on March 11, 2024, at Rizal Memorial Football Stadium. Scoring six goals, he earned the Golden Boot award and Rookie of the Year title. Kirk balances academics and sports, serving as an inspiration in both fields.

Appendix I: Documentation Pre-testing:





Intervention:







Post-Testing:



Appendix J: Statistician Certification

STATISTICIAN CERTIFICATION

This is to certify that this research study entitled "**THE EFFECTS OF PLYOMETRIC TRAINING ON SELECTED COLLEGIATE FOOTBALL PLAYERS' SPRINT SPEED, AGILITY AND FORCE GENERATION**" done by researchers; Mapula, Paul Vincent G., Pedimonte, Viron E., See, Kirk Eman R. in partial fulfilment of the requirements for the course Research 2 - Applied Research in Exercise & Sports has been statistically reviewed by the undersigned.

Dr. Ahli F Frigillana
Statistician
ORCID: 0000-0001-8615-0269